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Liu et al.

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- (54) **MEMORY CONTROLLER, MEMORY SYSTEM, AND METHOD FOR MANAGING LOGICAL-TO-PHYSICAL MAPPING TABLE BASED ON ADDRESS BOUNDARY**
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G06F 12/02 (2006.01)

(52) **U.S. Cl.**
CPC .. **G06F 12/0292** (2013.01); **G06F 2212/1024** (2013.01); **G06F 2212/2022** (2013.01); **G06F 2212/214** (2013.01); **G06F 2212/7201** (2013.01)

(58) **Field of Classification Search**
CPC **G06F 12/0292**; **G06F 2212/1024**; **G06F 2212/2022**; **G06F 2212/214**; **G06F 2212/7201**

See application file for complete search history.

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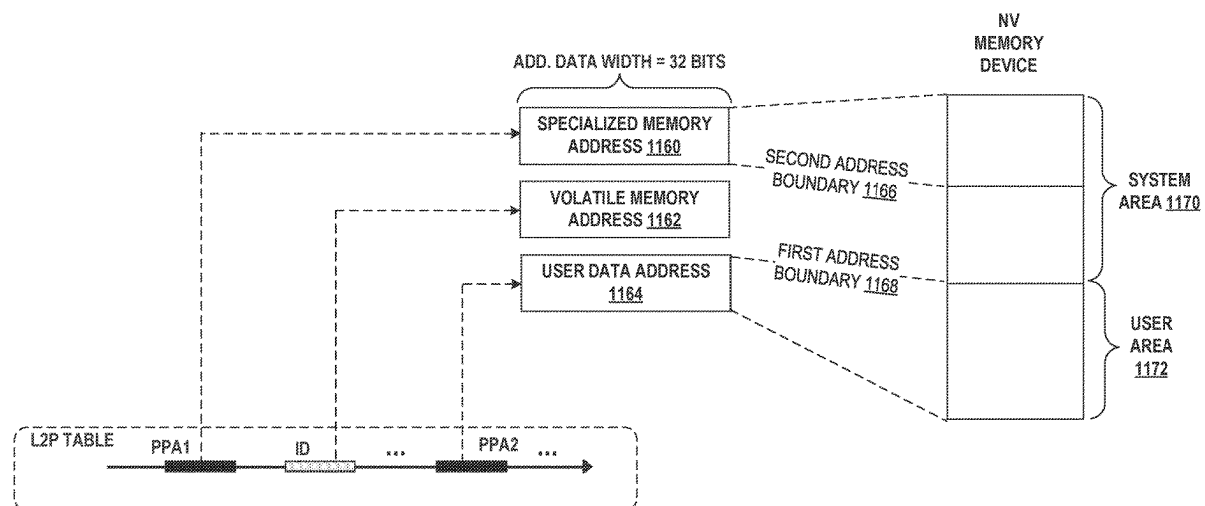
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(57) **ABSTRACT**

In certain aspects, a memory controller includes a logical-to-physical (L2P) search engine. The L2P search engine is configured to maintain an L2P mapping table that maps logical addresses to physical addresses, respectively. The L2P search engine is also configured to organize the physical addresses mapped by the L2P mapping table into address categories based on at least one address boundary.

20 Claims, 15 Drawing Sheets



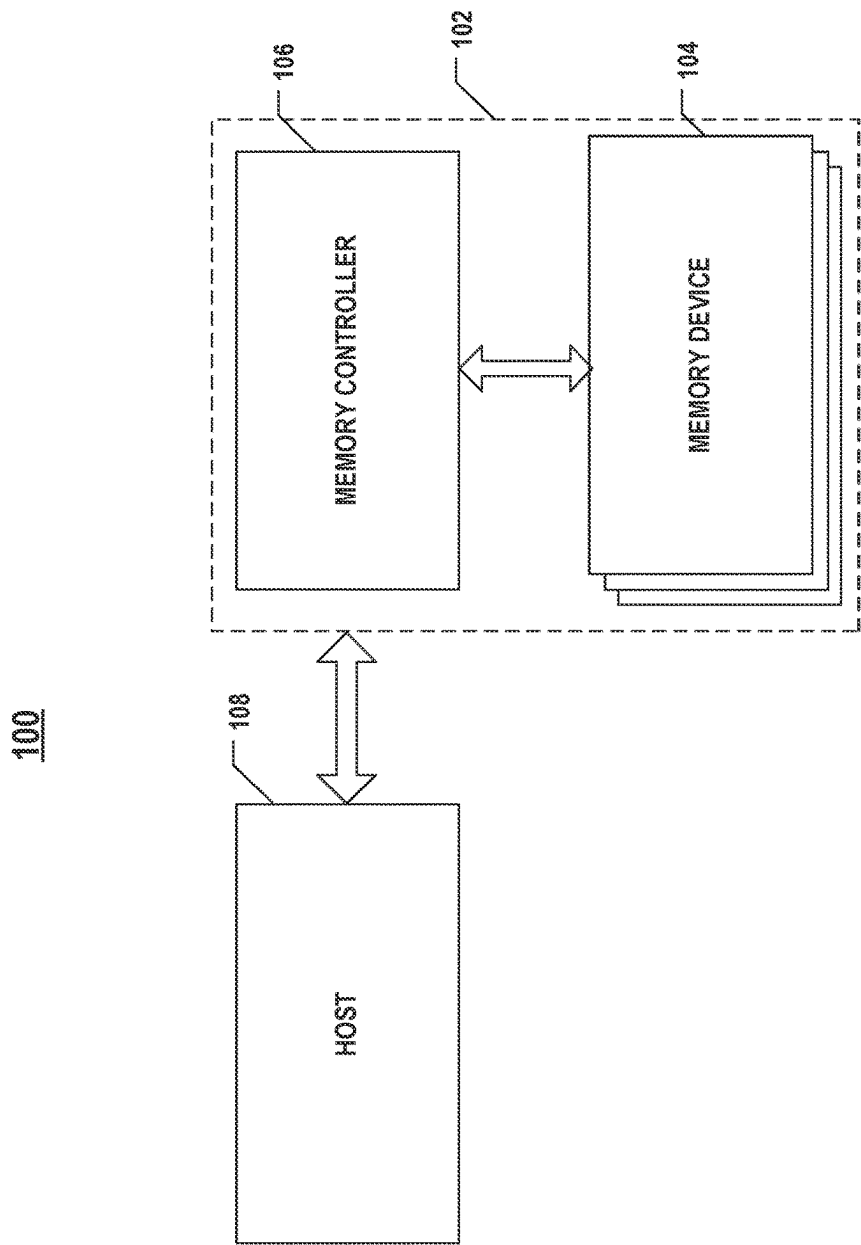


FIG. 1

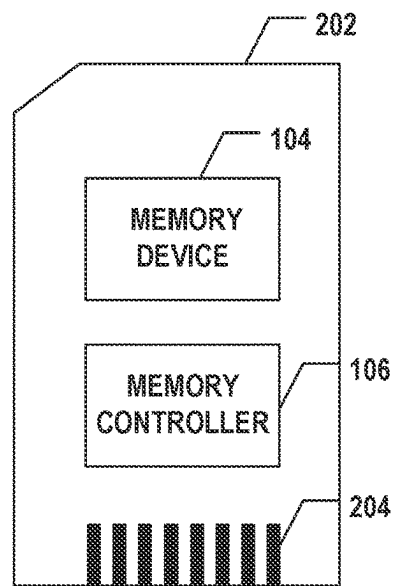


FIG. 2A

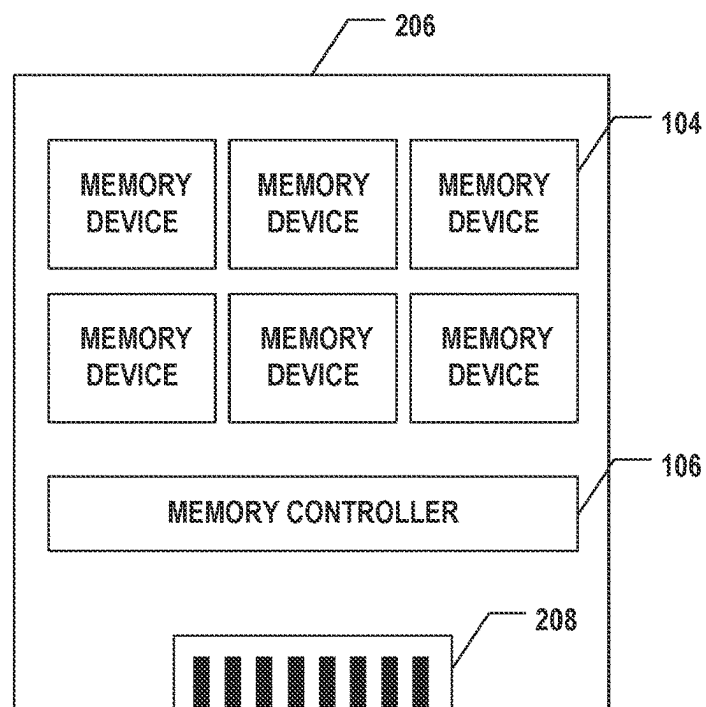


FIG. 2B

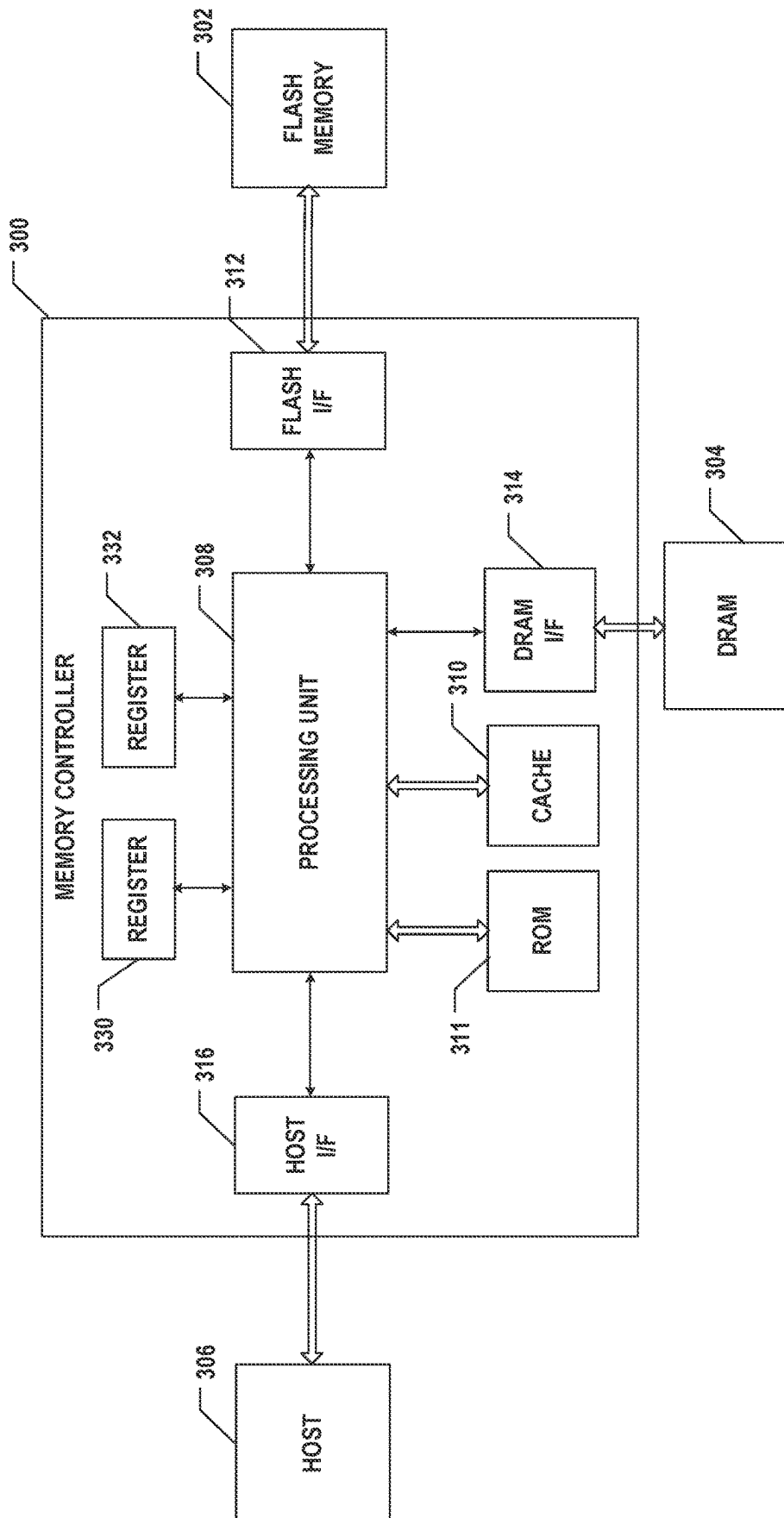


FIG. 3

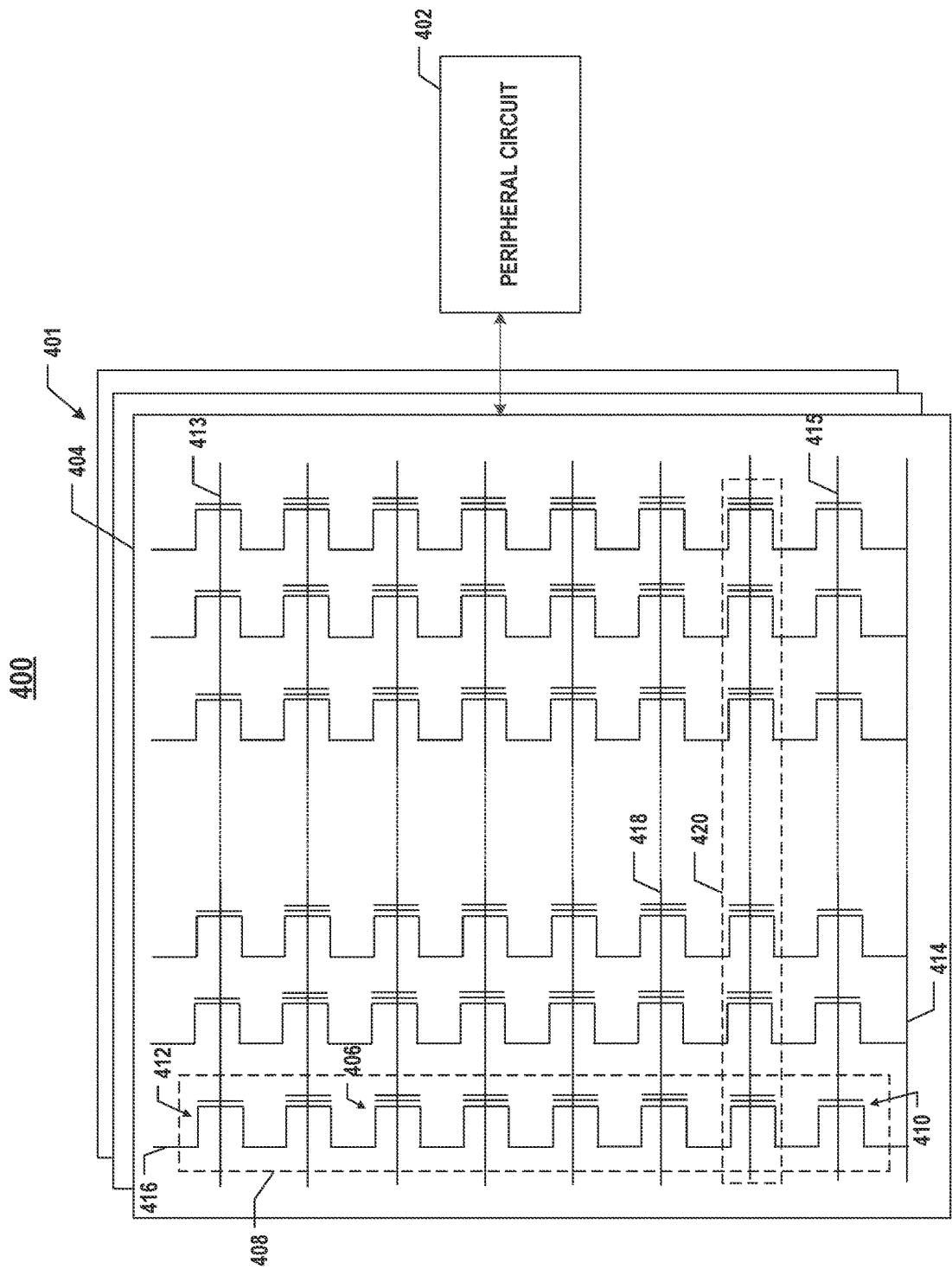


FIG. 4

500

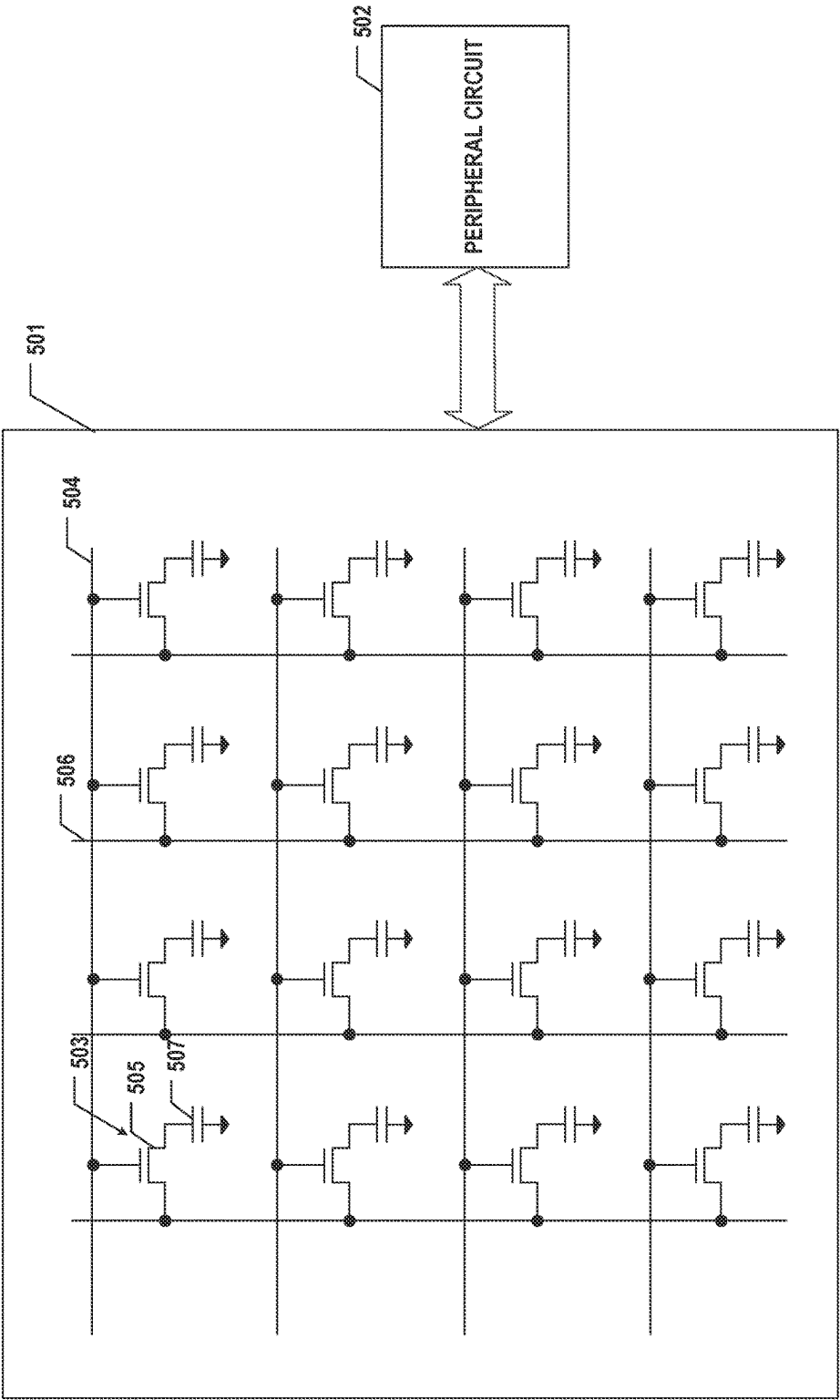


FIG. 5

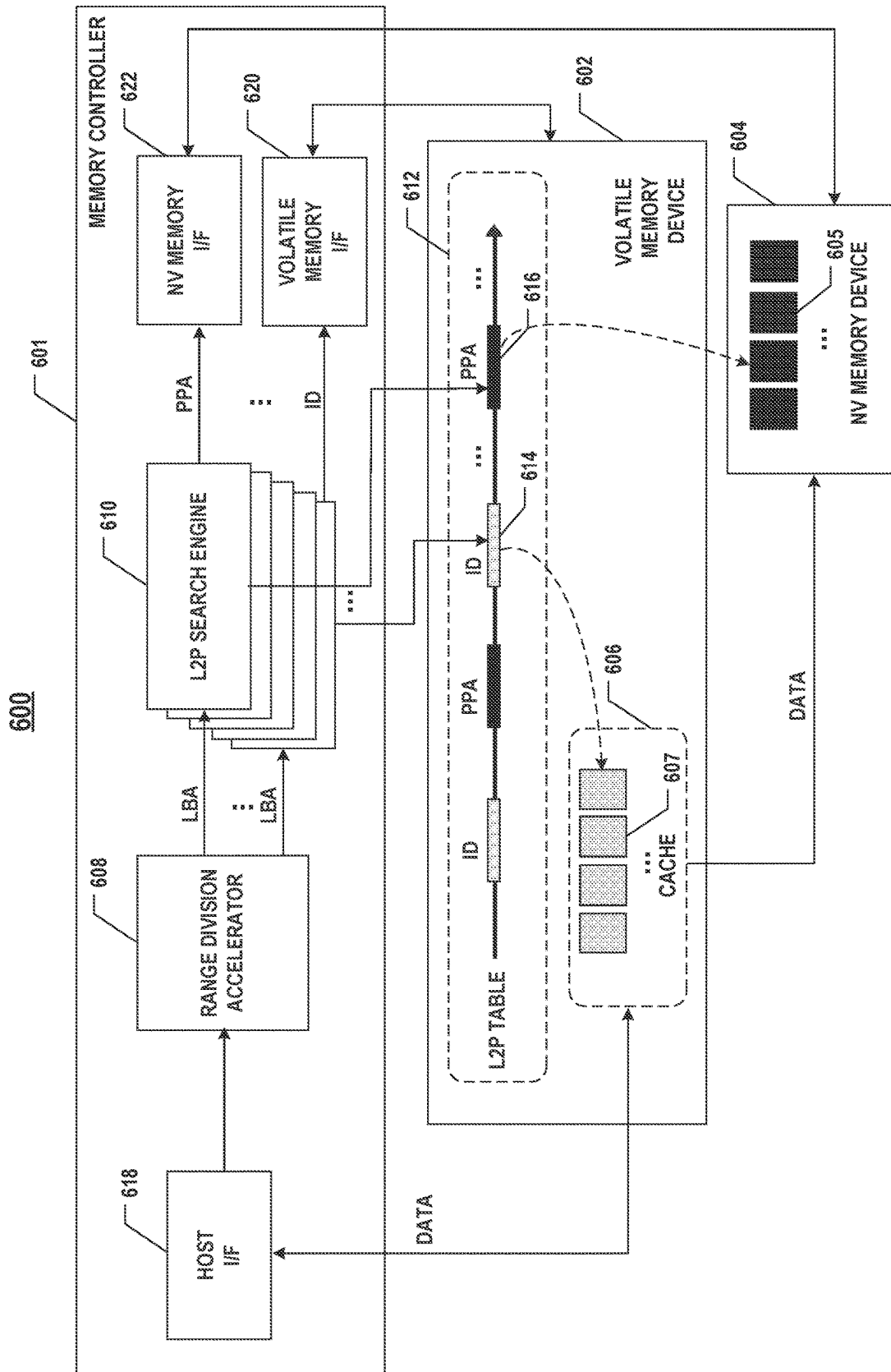


FIG. 6

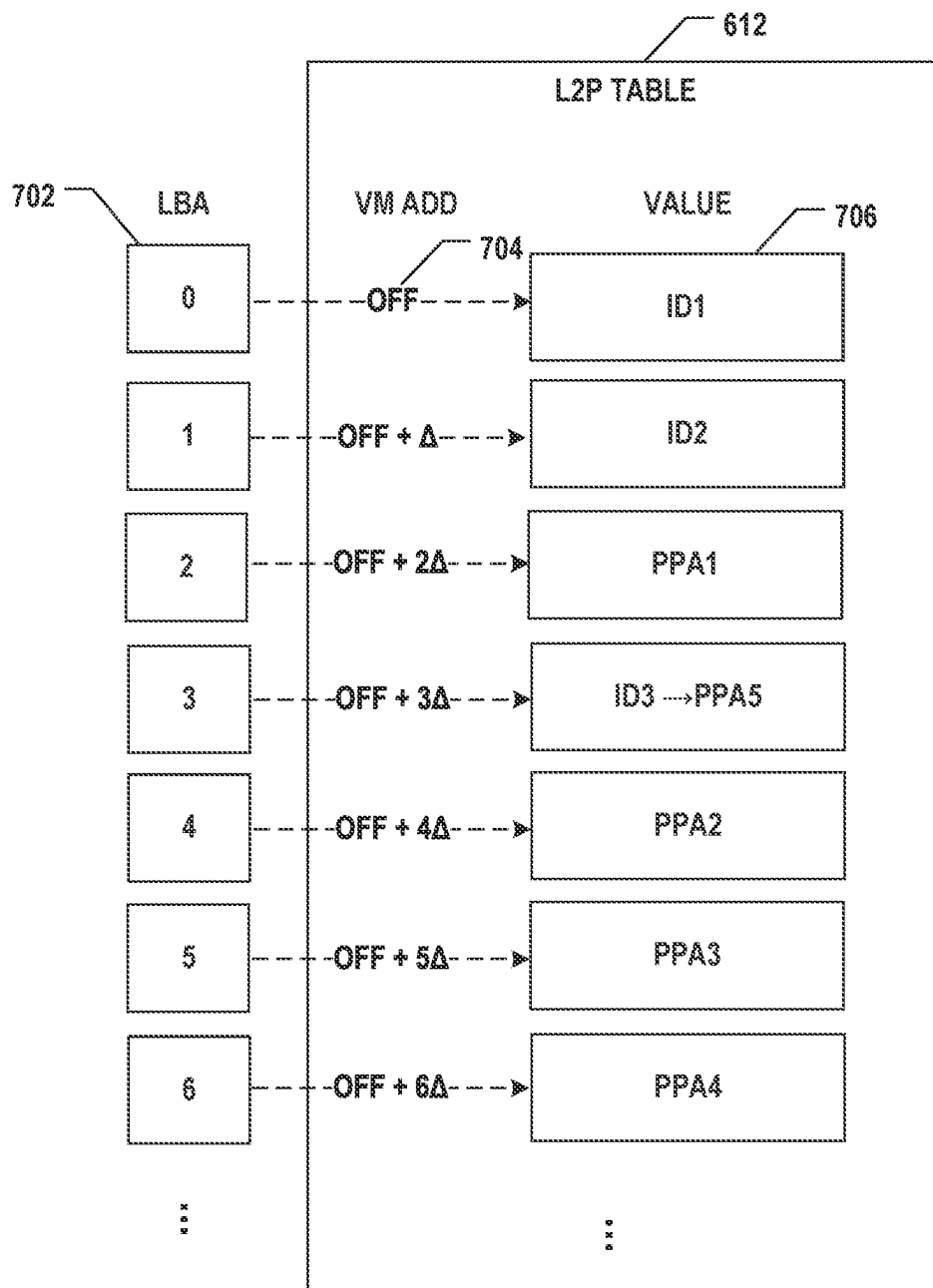


FIG. 7

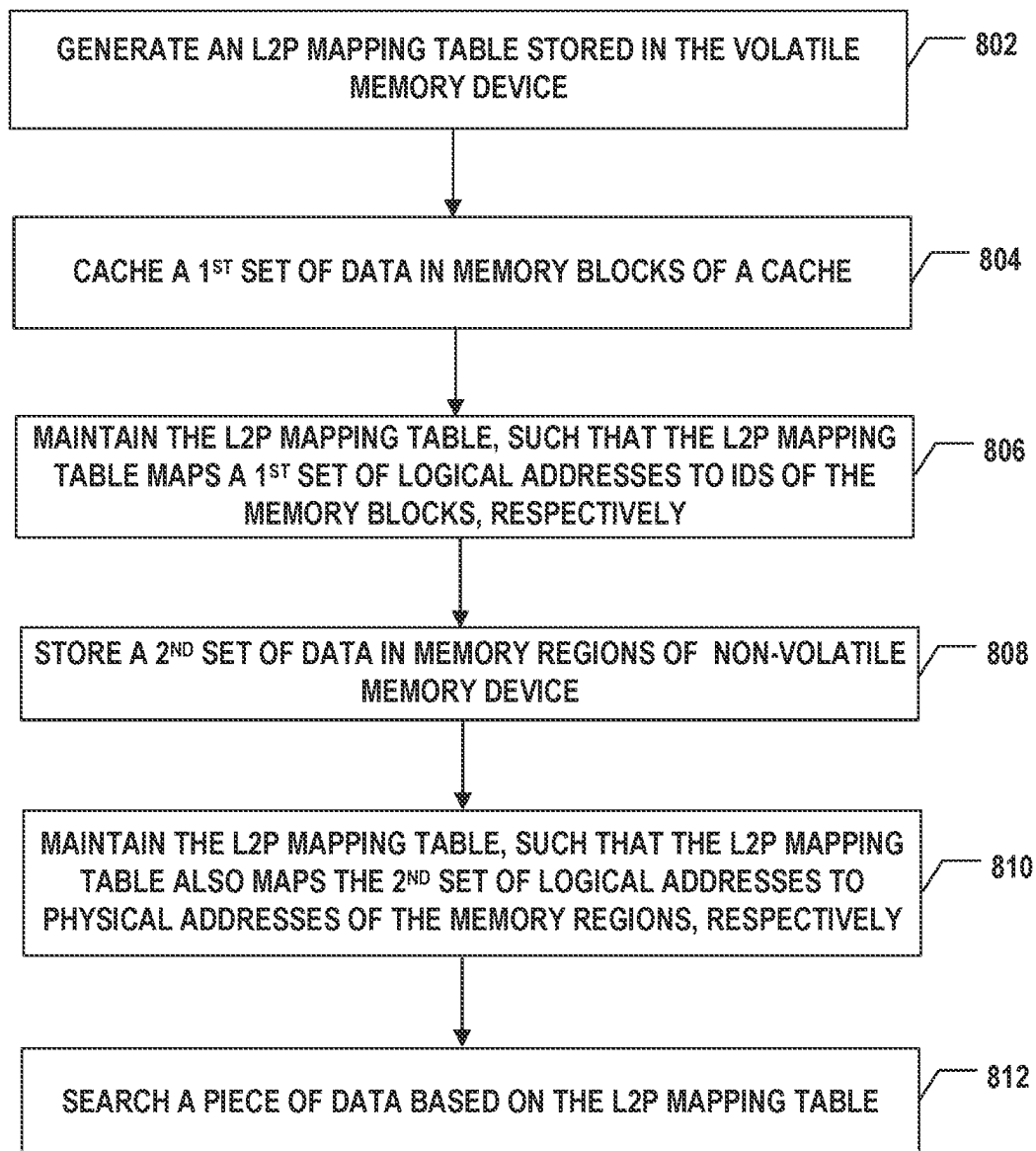
800

FIG. 8

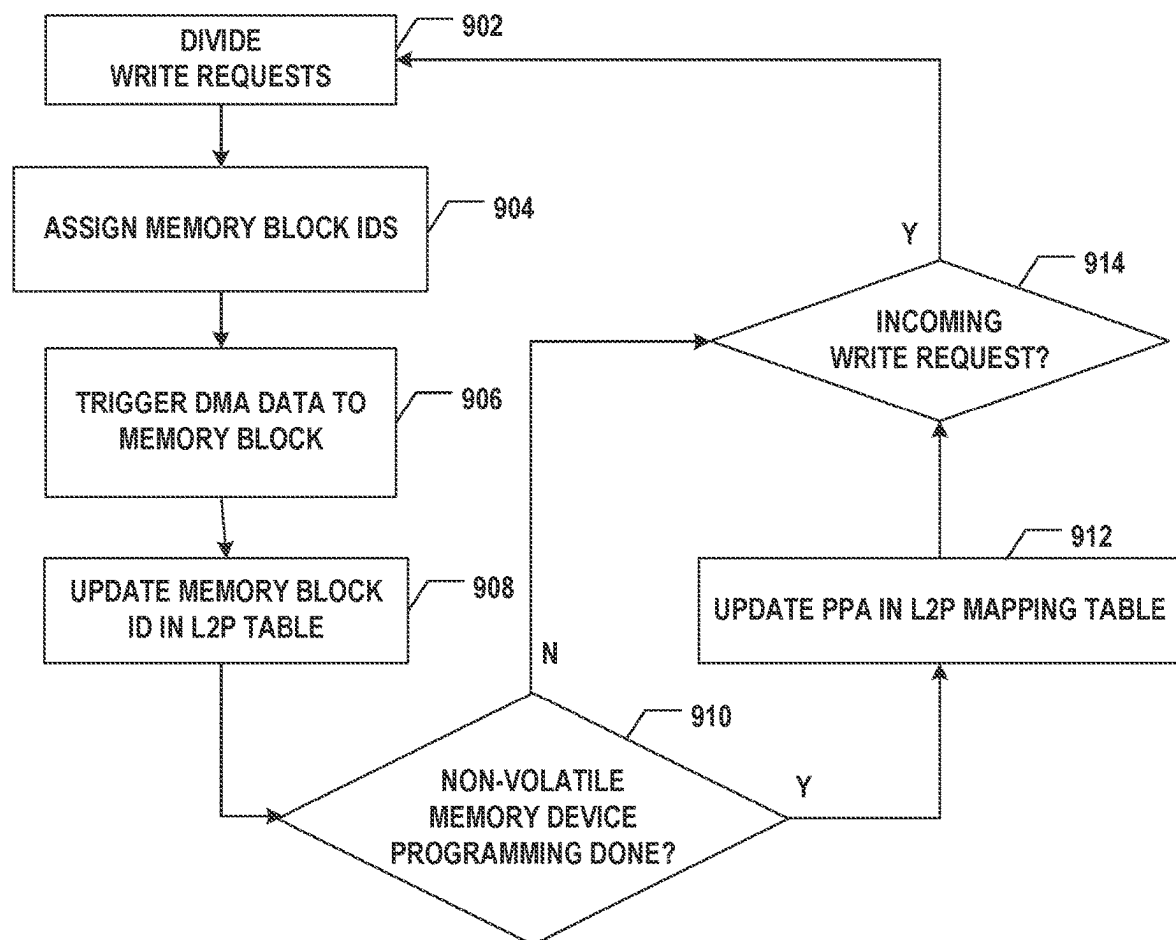


FIG. 9

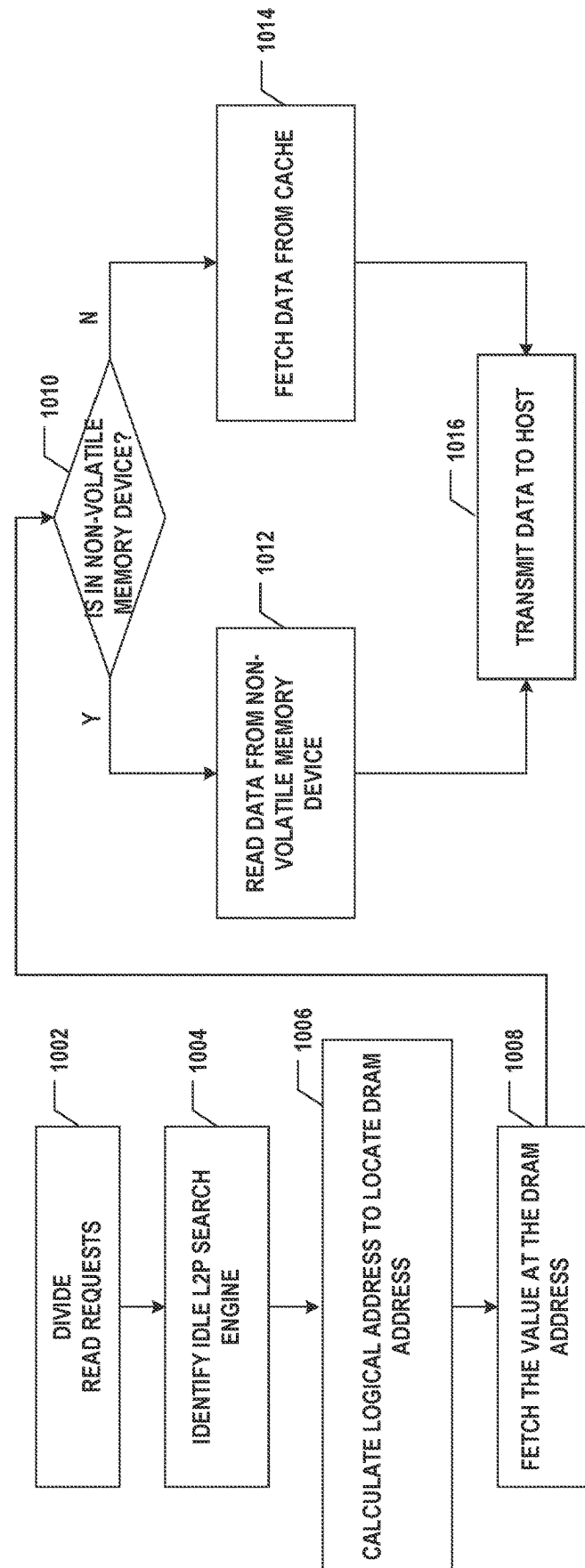
1000

FIG. 10

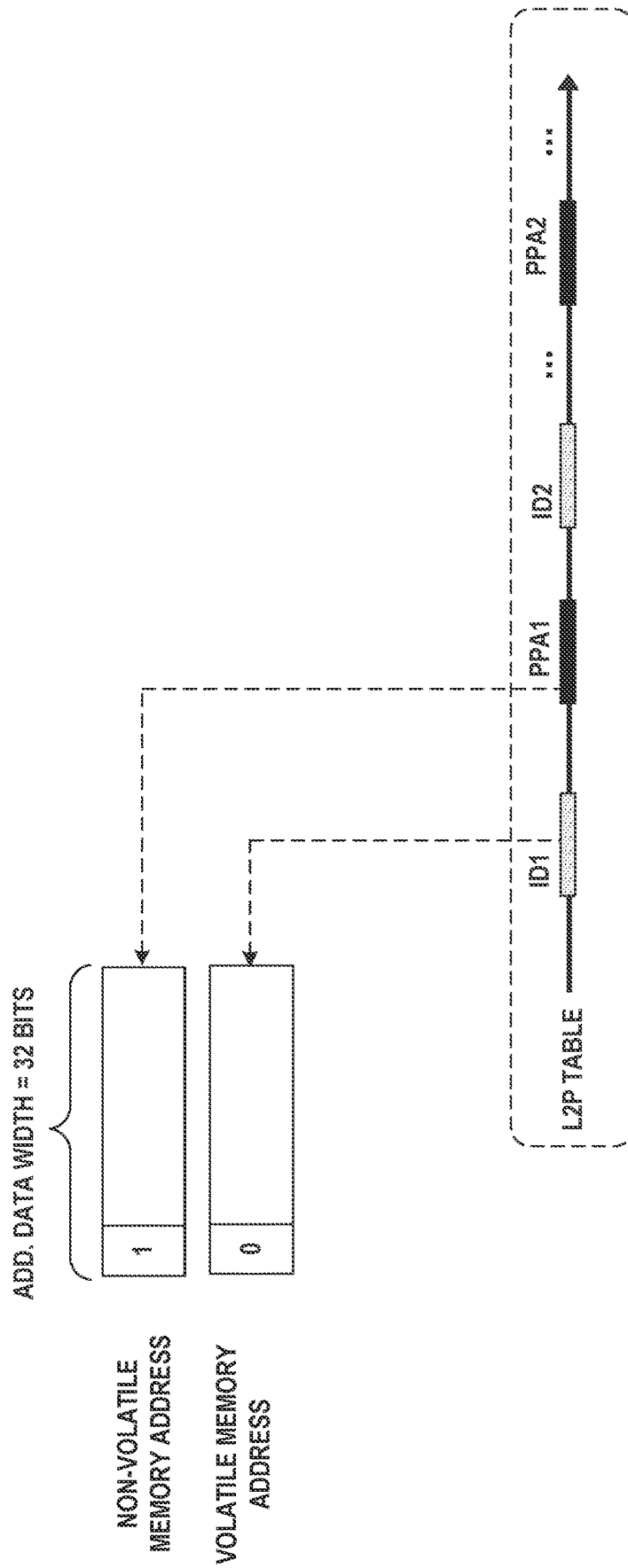


FIG. 11A

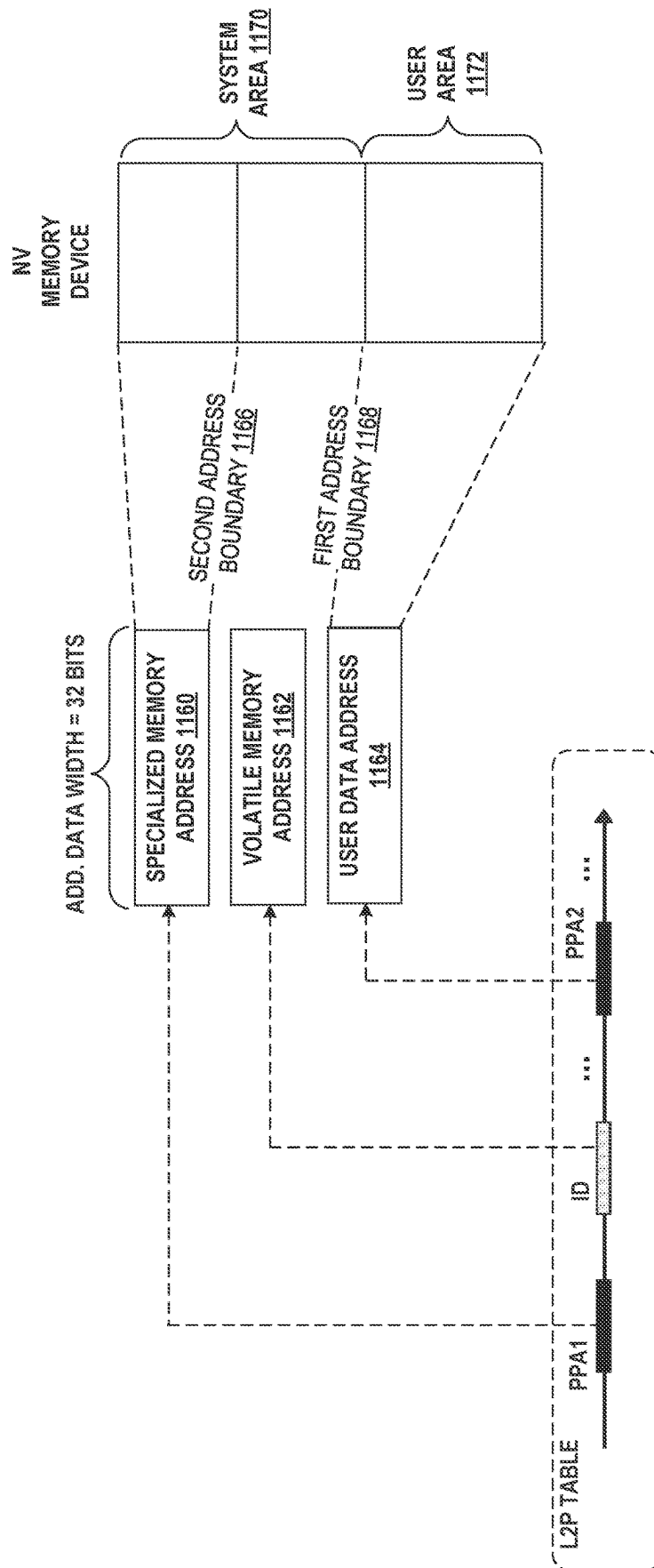


FIG. 11B

1200

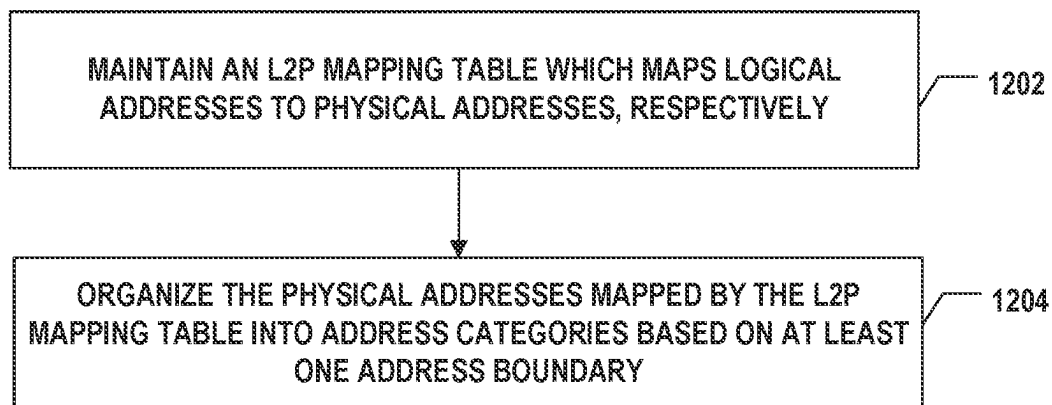


FIG. 12

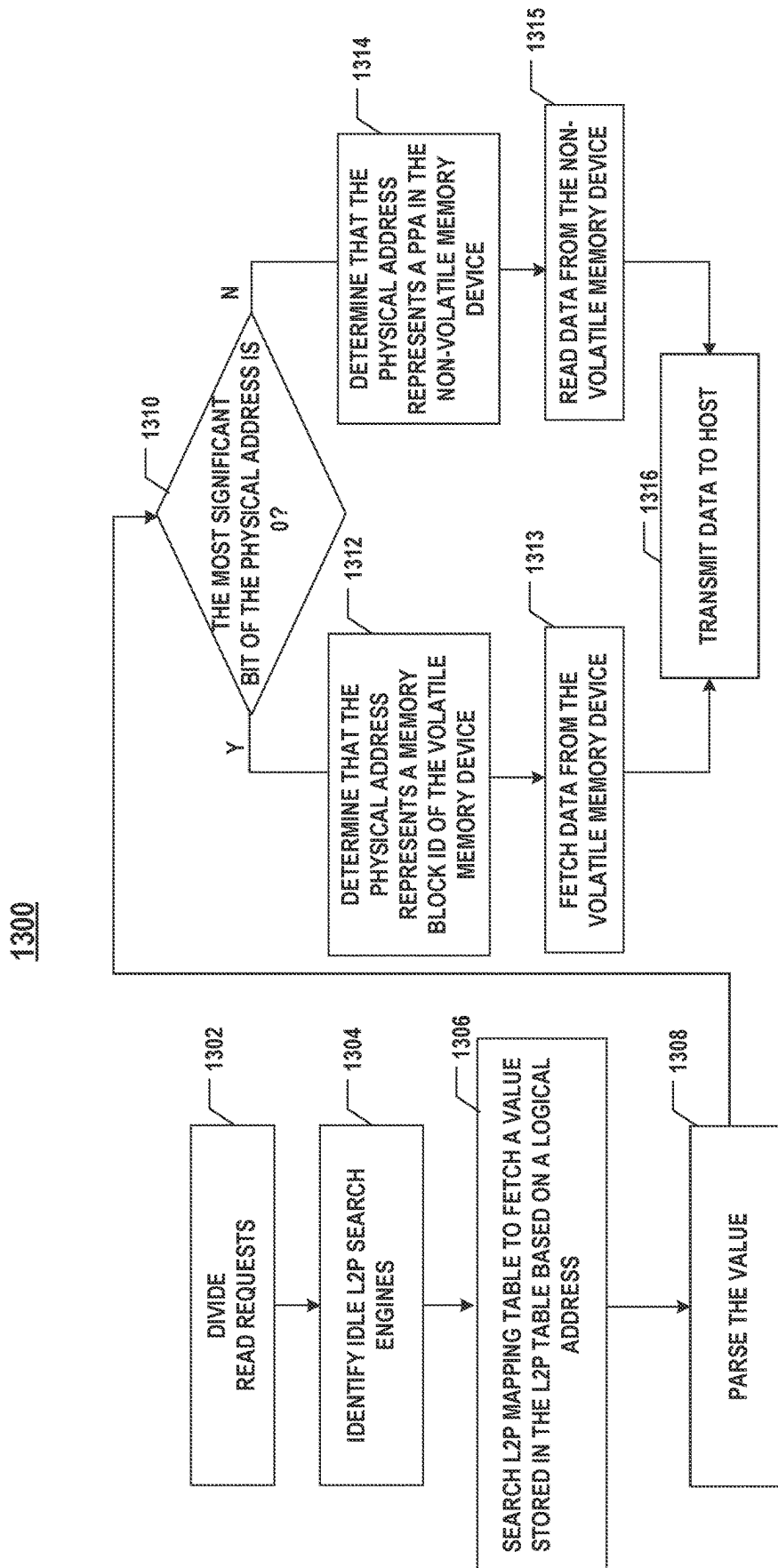


FIG. 13A

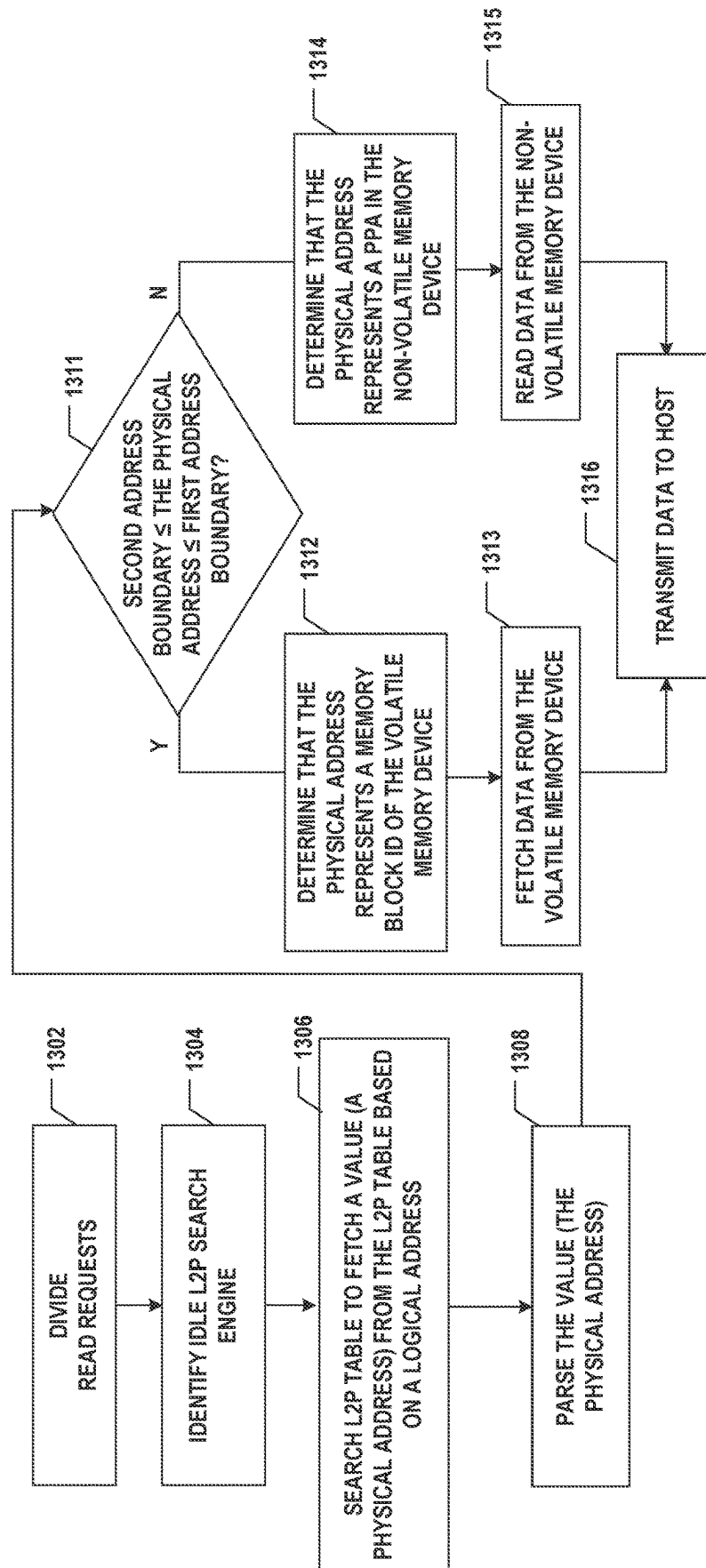
1350

FIG. 13B

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MEMORY CONTROLLER, MEMORY SYSTEM, AND METHOD FOR MANAGING LOGICAL-TO-PHYSICAL MAPPING TABLE BASED ON ADDRESS BOUNDARY

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of International Application No. PCT/CN2023/099321, filed on Jun. 9, 2023, entitled “MEMORY CONTROLLER, MEMORY SYSTEM MANAGING LOGICAL-TO-PHYSICAL MAPPING TABLE, METHOD, AND STORAGE MEDIUM THEREOF,” which is incorporated herein by reference in its entirety.

BACKGROUND

The present disclosure relates to memory devices and operation methods thereof.

Solid-state drives (SSDs) are a type of non-volatile data storage devices that have gained significant popularity in recent years due to their numerous advantages over traditional hard disk drives (HDDs), such as faster read and write speed, durability and reliability, reduced power consumption, silent operation, and smaller form factors. SSDs typically may use NAND Flash memory for non-volatile storage. Some SSDs, for example enterprise SSDs, also may use volatile memory (e.g., dynamic random-access memory (DRAM)) to enhance their performance, allowing faster access to data and more efficient handling of read and write operations.

SUMMARY

In one aspect, a memory controller includes a logical-to-physical (L2P) search engine. The L2P search engine is configured to maintain an L2P mapping table that maps logical addresses to physical addresses, respectively. The L2P search engine is further configured to organize the physical addresses mapped by the L2P mapping table into address categories based on at least one address boundary.

In some implementations, the at least one address boundary includes a first address boundary; and the memory controller further includes a first register configured to store the first address boundary.

In some implementations, the address categories include a first category of user data addresses mapping to memory regions of a user area of a non-volatile memory device and a second category of volatile memory addresses mapping to memory blocks of a volatile memory device.

In some implementations, the L2P mapping table maps a first set of logical addresses in the logical addresses to a first set of physical addresses associated with the memory regions of the user area of the non-volatile memory device, respectively, and the first category of user data addresses includes the first set of physical addresses associated with the memory regions of the user area of the non-volatile memory device. The L2P mapping table also maps a second set of logical addresses in the logical addresses to identifiers (IDs) of the memory blocks of the volatile memory device, respectively, and the second category of volatile memory addresses includes the IDs of the memory blocks of the volatile memory device.

In some implementations, each user data address in the first category is greater than the first address boundary; and

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each volatile memory address in the second category is smaller than the first address boundary.

In some implementations, the at least one address boundary further includes a second address boundary lower than the first address boundary; and the memory controller further includes a second register configured to store the second address boundary.

In some implementations, the address categories further include a third category of specialized memory addresses mapping to memory regions of a system area of the non-volatile memory device. The L2P mapping table also maps a third set of logical addresses in the logical addresses to a third set of physical addresses associated with the memory regions of the system area of the non-volatile memory device, respectively, and the third category of specialized memory addresses includes the third set of physical addresses associated with the memory regions of the system area of the non-volatile memory device.

In some implementations, each volatile memory address in the second category is greater than or equal to the second address boundary and smaller than or equal to the first address boundary; and each specialized memory address in the third category is smaller than the second address boundary.

In some implementations, the memory controller further includes a volatile memory device interface operatively coupled to the volatile memory device and a non-volatile memory device interface operatively coupled to the non-volatile memory device.

In some implementations, the volatile memory device includes dynamic random-access memory (DRAM), and the non-volatile memory device includes NAND Flash memory.

In some implementations, responsive to a read request indicative of retrieving a piece of data associated with a logical address, the L2P search engine is further configured to determine an address of an entry in the L2P mapping table based on the logical address, identify a physical address stored in the entry of the L2P mapping table based on the address of the entry, determine an address category which the physical address is classified into based on the first address boundary and the second address boundary, and instruct to fetch the piece of data from one of the volatile memory device and the non-volatile memory device based on the address category.

In some implementations, to determine the address category, the L2P search engine is further configured to, responsive to the physical address being greater than the first address boundary, determine that the physical address is classified into the first category of user data addresses mapping to the memory regions of the user area of the non-volatile memory device. Responsive to the physical address being lower than the second address boundary, the L2P search engine is further configured to determine that the physical address is classified into the third category of specialized memory addresses mapping to the memory regions of the system area of the non-volatile memory device. Or, responsive to the physical address being equal to or greater than the second address boundary and being equal to or smaller than the first address boundary, the L2P search engine is further configured to determine that the physical address is classified into the second category of volatile memory addresses mapping to the memory blocks of the volatile memory device.

In some implementations, to instruct to fetch the piece of data, the L2P search engine is further configured to, responsive to the physical address being classified into the third category of specialized memory addresses or the first cat-

egory of user data addresses, instruct to read the piece of data from the non-volatile memory device using the physical address. Or, responsive to the physical address being classified into the second category of volatile memory addresses, the L2P search engine is further configured to instruct to fetch the piece of data from the volatile memory device using the physical address.

In another aspect, a memory system includes a non-volatile memory device including memory regions each associated with a physical address and a memory controller operatively coupled to the non-volatile memory device. The memory controller is configured to control the non-volatile memory device. The memory controller includes an L2P search engine. The L2P search engine is configured to maintain an L2P mapping table that maps logical addresses to physical addresses, respectively, and organize the physical addresses mapped by the L2P mapping table into address categories based on at least one address boundary.

In some implementations, the at least one address boundary includes a first address boundary; and the memory controller further includes a first register configured to store the first address boundary.

In some implementations, the address categories include a first category of user data addresses mapping to memory regions of a user area of a non-volatile memory device and a second category of volatile memory addresses mapping to memory blocks of a volatile memory device.

In some implementations, the L2P mapping table maps a first set of logical addresses in the logical addresses to a first set of physical addresses associated with the memory regions of the user area of the non-volatile memory device, respectively, and the first category of user data addresses includes the first set of physical addresses associated with the memory regions of the user area of the non-volatile memory device. The L2P mapping table also maps a second set of logical addresses in the logical addresses to IDs of the memory blocks of the volatile memory device, respectively, and the second category of volatile memory addresses includes the IDs of the memory blocks of the volatile memory device.

In some implementations, each user data address in the first category is greater than the first address boundary; and each volatile memory address in the second category is smaller than the first address boundary.

In some implementations, the L2P mapping table is stored in the volatile memory device.

In some implementations, the at least one address boundary further includes a second address boundary lower than the first address boundary; and the memory controller further includes a second register configured to store the second address boundary.

In some implementations, the address categories further include a third category of specialized memory addresses mapping to memory regions of a system area of the non-volatile memory device. The L2P mapping table also maps a third set of logical addresses in the logical addresses to a third set of physical addresses associated with the memory regions of the system area of the non-volatile memory device, respectively, and the third category of specialized memory addresses includes the third set of physical addresses associated with the memory regions of the system area of the non-volatile memory device.

In some implementations, each volatile memory address in the second category is greater than or equal to the second address boundary and smaller than or equal to the first

address boundary; and each specialized memory address in the third category is smaller than the second address boundary.

In some implementations, the memory controller further includes a volatile memory device interface operatively coupled to the volatile memory device and a non-volatile memory device interface operatively coupled to the non-volatile memory device.

In some implementations, the volatile memory device includes DRAM, and the non-volatile memory device includes NAND Flash memory.

In some implementations, responsive to a read request indicative of retrieving a piece of data associated with a logical address, the L2P search engine is further configured to determine an address of an entry in the L2P mapping table based on the logical address, identify a physical address stored in the entry of the L2P mapping table based on the address of the entry, determine an address category which the physical address is classified into based on the first address boundary and the second address boundary, and instruct to fetch the piece of data from one of the volatile memory device and the non-volatile memory device based on the address category.

In some implementations, to determine the address category, the L2P search engine is further configured to, responsive to the physical address being greater than the first address boundary, determine that the physical address is classified into the first category of user data addresses mapping to the memory regions of the user area of the non-volatile memory device. Responsive to the physical address being lower than the second address boundary, the L2P search engine is further configured to determine that the physical address is classified into the third category of specialized memory addresses mapping to the memory regions of the system area of the non-volatile memory device. Or, responsive to the physical address being equal to or greater than the second address boundary and being equal to or smaller than the first address boundary, the L2P search engine is further configured to determine that the physical address is classified into the second category of volatile memory addresses mapping to the memory blocks of the volatile memory device.

In some implementations, to instruct to fetch the piece of data, the L2P search engine is further configured to, responsive to the physical address being classified into the third category of specialized memory addresses or the first category of user data addresses, instruct to read the piece of data from the non-volatile memory device using the physical address. Or, responsive to the physical address being classified into the second category of volatile memory addresses, the L2P search engine is further configured to instruct to fetch the piece of data from the volatile memory device using the physical address.

In still another aspect, a method for operating a memory controller is provided. An L2P mapping table that maps logical addresses to physical addresses, respectively, is maintained. The physical addresses mapped by the L2P mapping table are organized into address categories based on at least one address boundary.

In some implementations, the address boundaries include a first address boundary stored in a first register.

In some implementations, the address categories include a first category of user data addresses mapping to memory regions of a user area of a non-volatile memory device and a second category of volatile memory addresses mapping to memory blocks of a volatile memory device.

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In some implementations, the L2P mapping table maps a first set of logical addresses in the logical addresses to a first set of physical addresses associated with the memory regions of the user area of the non-volatile memory device, respectively, and the first category of user data addresses includes the first set of physical addresses associated with the memory regions of the user area of the non-volatile memory device. The L2P mapping table also maps a second set of logical addresses in the logical addresses to IDs of the memory blocks of the volatile memory device, respectively, and the second category of volatile memory addresses includes the IDs of the memory blocks of the volatile memory device.

In some implementations, each user data address in the first category is greater than the first address boundary; and each volatile memory address in the second category is smaller than the first address boundary.

In some implementations, the at least one address boundary further includes a second address boundary lower than the first address boundary.

In some implementations, the address categories further include a third category of specialized memory addresses mapping to memory regions of a system area of the non-volatile memory device. The L2P mapping table also maps a third set of logical addresses in the logical addresses to a third set of physical addresses associated with the memory regions of the system area of the non-volatile memory device, respectively, and the third category of specialized memory addresses includes the third set of physical addresses associated with the memory regions of the system area of the non-volatile memory device.

In some implementations, each volatile memory address in the second category is greater than or equal to the second address boundary and smaller than or equal to the first address boundary; and each specialized memory address in the third category is smaller than the second address boundary.

In some implementations, the volatile memory device includes DRAM, and the non-volatile memory device includes NAND Flash memory.

In some implementations, responsive to a read request indicative of retrieving a piece of data associated with a logical address, an address of an entry in the L2P mapping table based on the logical address is determined. A physical address stored in the entry of the L2P mapping table is identified based on the address of the entry. An address category which the physical address is classified into is determined based on the first address boundary and the second address boundary. To fetch the piece of data from one of the volatile memory device and the non-volatile memory device is instructed based on the address category.

In some implementations, determining the address category includes, responsive to the physical address being greater than the first address boundary, determining that the physical address is classified into the first category of user data addresses mapping to the memory regions of the user area of the non-volatile memory device. Responsive to the physical address being lower than the second address boundary, determining the address category includes determining that the physical address is classified into the third category of specialized memory addresses mapping to the memory regions of the system area of the non-volatile memory device. Or, responsive to the physical address being equal to or greater than the second address boundary and being equal to or smaller than the first address boundary, determining the address category includes determining that the physical

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address is classified into the second category of volatile memory addresses mapping to the memory blocks of the volatile memory device.

In some implementations, instructing to fetch the piece of data includes: responsive to the physical address being classified into the third category of specialized memory addresses or the first category of user data addresses, instructing to read the piece of data from the non-volatile memory device using the physical address; or responsive to the physical address being classified into the second category of volatile memory addresses, instructing to fetch the piece of data from the volatile memory device using the physical address.

In yet another aspect, a non-transitory computer-readable storage medium storing instructions is disclosed. The instructions, when executed by a memory controller of a memory system, cause the memory controller to perform a method. The method includes maintaining an L2P mapping table which maps logical addresses to physical addresses, respectively. The method also includes organizing the physical addresses mapped by the L2P mapping table into address categories based on at least one address boundary.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are incorporated herein and form a part of the specification, illustrate aspects of the present disclosure and, together with the description, further serve to explain the principles of the present disclosure and to enable a person skilled in the pertinent art to make and use the present disclosure.

FIG. 1 illustrates a block diagram of a system including a memory system, according to some aspects of the present disclosure.

FIG. 2A illustrates a diagram of a memory card having a memory device, according to some aspects of the present disclosure.

FIG. 2B illustrates a diagram of an SSD having memory devices, according to some aspects of the present disclosure.

FIG. 3 illustrates a block diagram of a memory controller, according to some aspects of the present disclosure.

FIG. 4 illustrates a schematic diagram of a NAND Flash memory device including peripheral circuits, according to some aspects of the present disclosure.

FIG. 5 illustrates a schematic diagram of a DRAM device including peripheral circuits, according to some aspects of the present disclosure.

FIG. 6 illustrates a detailed schematic diagram of a memory system managing an L2P mapping table and performing data search based on the L2P mapping table, according to some aspects of the present disclosure.

FIG. 7 illustrates an L2P mapping table, according to some aspects of the present disclosure.

FIG. 8 illustrates a flowchart of a method for operating a memory controller, according to some aspects of the present disclosure.

FIG. 9 illustrates a flowchart of a method for handling write requests, according to some aspects of the present disclosure.

FIG. 10 illustrates a flowchart of a method for handling read requests, according to some aspects of the present disclosure.

FIG. 11A illustrates an example to organize physical addresses mapped by an L2P mapping table, according to some aspects of the present disclosure.

FIG. 11B illustrates another example to organize physical addresses mapped by an L2P mapping table, according to some aspects of the present disclosure.

FIG. 12 illustrates a flowchart of another method for operating a memory controller, according to some aspects of the present disclosure.

FIG. 13A illustrates a flowchart of an implementation of a method for handling a read request when physical addresses mapped by an L2P mapping table are organized using the example approach of FIG. 11A, according to some aspects of the present disclosure.

FIG. 13B illustrates a flowchart of an implementation of another method for handling a read request when physical addresses mapped by an L2P mapping table are organized using the example approach of FIG. 11B, according to some aspects of the present disclosure.

The present disclosure will be described with reference to the accompanying drawings.

DETAILED DESCRIPTION

In general, terminology may be understood at least in part from usage in context. For example, the term “one or more” as used herein, depending at least in part upon context, may be used to describe any feature, structure, or characteristic in a singular sense or may be used to describe combinations of features, structures or characteristics in a plural sense. Similarly, terms, such as “a,” “an,” or “the,” again, may be understood to convey a singular usage or to convey a plural usage, depending at least in part upon context. In addition, the term “based on” may be understood as not necessarily intended to convey an exclusive set of factors and may, instead, allow for existence of additional factors not necessarily expressly described, again, depending at least in part on context.

In an SSD scenario (e.g., enterprise SSD), a mapping relationship between logical addresses and physical addresses is recorded by an L2P mapping table which can be stored on a DRAM of the SSD for data tracking purposes. The logical addresses (such as logical block addresses (LBAs)) can be used as indices of the various entries of the L2P mapping table. The content of each entry of the L2P mapping table can be a physical address (such as a physical page address (PPA)) corresponding to the logical address of the entry. In some examples, the size of the L2P mapping table is equal to 1/1024 of the drive capacity of the SSD, which is large and may occupy a significant amount of storage space of the DRAM. The ratio between the size of the L2P mapping table and the drive capacity of the SSD is 1/1024 because the L2P mapping table may use an address data width of 4 bytes to express 4 KiB user data on the SSD. For an SSD with a small capacity, the address data width of 4 bytes can be sufficient to express physical addresses of corresponding physical space of the SSD. However, for an SSD (e.g., enterprise SSD) with a large capacity (e.g., 2 TB, 4 TB, etc.), the address data width of 4 bytes may be insufficient to express the physical addresses of corresponding physical space of the SSD, especially when part of the 4 bytes (e.g., one of the 32 bits) is reserved for marking the types of the physical addresses.

To address one or more of the aforementioned issues, the present disclosure introduces an address management scheme for an L2P mapping table, which does not need to reserve any bits for marking the type or purpose of a physical address. For example, the L2P mapping table may map a plurality of logical addresses to a plurality of physical addresses. The address management scheme disclosed

herein can organize the plurality of physical addresses mapped by the L2P mapping table into a plurality of address categories based on at least one address boundary (e.g., a first address boundary and a second address boundary). The plurality of address categories may include at least one of (1) a first category of user data addresses mapping to memory regions of a user area of a non-volatile memory device, (2) a second category of volatile memory addresses (e.g., IDs of memory blocks of a cache or DRAM), or (3) a third category of specialized memory addresses mapping to memory regions of a system area of a non-volatile memory device. No bits are needed to be reserved for distinguishing the different categories of the physical addresses because the first address boundary and the second address boundary can be used to determine the categories of the physical addresses. As a result, all bits in the address data width (such as 32 bits) can be used to express a larger physical space. The size of the L2P mapping table can be reduced, and thus, the size of the DRAM in the enterprise SSD can also be reduced, leading to a reduction in the cost of the DRAM.

FIG. 1 illustrates a block diagram of a system 100 including a memory system 102, according to some aspects of the present disclosure. System 100 can be a mobile phone, a desktop computer, a laptop computer, a tablet, a vehicle computer, a gaming console, a printer, a positioning device, a wearable electronic device, a smart sensor, a virtual reality (VR) device, an argument reality (AR) device, or any other suitable electronic devices having storage therein. As shown in FIG. 1, system 100 can include a host 108 and memory system 102 having one or more memory devices 104 and a memory controller 106. Host 108 can be a processor of an electronic device, such as a central processing unit (CPU), or a system-on-chip (SoC), such as an application processor (AP). Host 108 can be configured to send or receive data (a.k.a. user data or host data) to or from memory system 102. Memory system 102 can be a storage product integrating memory controller 106 and one or more memory devices 104, such as an SSD.

Memory devices 104 can be any memory devices disclosed in the present disclosure, including non-volatile memory devices, such as NAND Flash memory devices. In some implementations, memory device 104 also includes one or more volatile memory devices, such as DRAM devices or static random-access memory (SRAM) devices.

Memory controller 106 is operatively coupled to memory devices 104 and host 108 and is configured to control memory devices 104, according to some implementations. Memory controller 106 can manage the data stored in memory devices 104 and communicate with host 108. In some implementations, memory controller 106 is designed for operating in a low duty-cycle environment like secure digital (SD) cards, compact Flash (CF) cards, universal serial bus (USB) Flash drives, or other media for use in electronic devices, such as personal computers, digital cameras, mobile phones, etc. In some implementations, memory controller 106 is designed for operating in a high duty-cycle environment with SSDs or embedded multimedia card (eMMCs) used as data storage for mobile devices, such as smartphones, tablets, laptop computers, etc., and enterprise storage arrays. Memory controller 106 can be configured to control operations of memory devices 104, such as read, program/write, and/or erase operations. Memory controller 106 can also be configured to manage various functions with respect to the data stored or to be stored in memory devices 104 including, but not limited to bad-block management, garbage collection, L2P address conversion, wear-leveling, etc. In some implementations, memory controller 106 is

further configured to process error correction codes (ECCs) with respect to the data read from or written to memory devices **104**. Any other suitable functions may be performed by memory controller **106** as well, for example, formatting memory devices **104**. Memory controller **106** can communicate with an external device (e.g., host **108**) according to a particular communication protocol. For example, memory controller **106** may communicate with the external device through at least one of various interface protocols, such as a non-volatile memory express (NVMe) protocol, an NVMe-over-fabrics (NVMe-oF) protocol, a PCI-express (PCI-E) protocol, a universal serial bus (USB) protocol, a multimedia card (MMC) protocol, a peripheral component interconnection (PCI) protocol, an advanced technology attachment (ATA) protocol, a serial-ATA protocol, a parallel-ATA protocol, a small computer small interface (SCSI) protocol, an enhanced small disk interface (ESDI) protocol, an integrated drive electronics (IDE) protocol, a Firewire protocol, etc.

Memory controller **106** and one or more memory devices **104** can be integrated into various types of storage devices, for example, being included in the same package, such as a universal Flash storage (UFS) package or an eMMC package. That is, memory system **102** can be implemented and packaged into different types of end electronic products. In one example as shown in FIG. 2A, memory controller **106** and a single memory device **104** may be integrated into a memory card **202**. Memory card **202** can include a PC card (PCMCIA, personal computer memory card international association), a CF card, a smart media (SM) card, a memory stick, a multimedia card (MMC, RS-MMC, MMCmicro), an SD card (SD, miniSD, microSD, SDHC), a UFS, etc. Memory card **202** can further include a memory card connector **204** coupling memory card **202** with a host (e.g., host **108** in FIG. 1). In another example as shown in FIG. 2B, memory controller **106** and multiple memory devices **104** may be integrated into an SSD **206**. SSD **206** can further include an SSD connector **208** coupling SSD **206** with a host (e.g., host **108** in FIG. 1). In some implementations, the storage capacity and/or the operation speed of SSD **206** is greater than those of memory card **202**. In some implementations, memory system **102** is implemented as an SSD **206** that includes both non-volatile memory devices and volatile memory devices as memory devices **104**, such as an enterprise SSD.

FIG. 3 illustrates a block diagram of a memory controller **300**, according to some aspects of the present disclosure. Memory controller **300** may be one example of memory controller **106** in FIG. 1. As shown in FIG. 3, memory controller **300** can include a processing unit **308**, a cache **310**, and a read-only memory (ROM) **311**. In some implementations, processing unit **308** is implemented by microprocessors (e.g., digital signal processors (DSPs)) or microcontrollers (a.k.a. microcontroller units (MCUs)) that execute firmware and/or software modules to perform the various functions described herein. The various firmware modules in memory controller **300** described herein can be implemented as firmware codes or instructions stored in ROM **311** and executed by processing unit **308**. In some implementations, processing unit **308** includes one or more hardware circuits, for example, fixed logic units such as a logic gate, a multiplexer, a flip-flop, a state machine, field-programmable gate arrays (FPGAs), programmable logic devices (PLDs). For example, the hardware circuits may include dedicated circuits performing a given logic function that is known at the time of device manufacture, such as application-specific integrated circuits (ASICs).

As shown in FIG. 3, memory controller **300** can also include various input/output (I/O) interfaces (I/F), such as a flash interface **312**, a DRAM interface **314**, and a host interface **316** operatively coupled to flash memory **302** (e.g., an example of non-volatile memory devices), DRAM **304** (e.g., an example of volatile memory devices), and a host **306** (e.g., an example of host **108**), respectively. flash interface **312**, DRAM interface **314**, and host interface **316** can be configured to transfer data, command, clock, or any suitable signals between processing unit **308** and flash memory **302**, DRAM **304**, and host **306**, respectively. flash interface **312**, DRAM interface **314**, and host interface **316** can implement any suitable communication protocols facilitating data transfer, communication, and management, such as the NVMe protocol and PCI-E protocol, double data rate (DDR) protocol, to name a few.

As described above, both cache **310** and DRAM **304** may be considered as volatile memory devices that can be controlled and accessed by memory controller **300** in a memory system. Consistent with the scope of the present disclosure, a cache can be implemented as part of volatile memory devices, for example, by cache **310** and/or DRAM **304**. It is understood that although FIG. 3 shows that cache **310** is within memory controller **300** and DRAM **304** is outside of memory controller **300**, in some examples, both cache **310** and DRAM **304** may be within memory controller **300** or outside of memory controller **300**.

Consistent with the scope of the present disclosure and disclosed below in detail, memory controller **300** can be configured to maintain an L2P mapping table that maps a plurality of logical addresses to a plurality of physical addresses, respectively. Memory controller **300** can also be configured to organize the plurality of physical addresses mapped by the L2P mapping table into a plurality of address categories based on at least one address boundary. In some implementations, memory controller **300** may further include at least one register configured to store the at least one address boundary. For example, the at least one address boundary may include a first address boundary and a second address boundary, and the at least one register may include registers **330** and **332** configured to store the first address boundary and the second address boundary, respectively. In some other implementations, there may be no registers in memory controller **300**, and memory controller **300** may receive the address boundaries from host **306** via host interface **316**. In some other implementations, the at least one address boundary may be stored in firmware of memory controller **300**, so that memory controller **300** does not need to retrieve the at least one address boundary from any hardware devices. Memory controller **300** is described below in more detail with reference to FIGS. 6-13B.

FIG. 4 illustrates a schematic circuit diagram of a NAND Flash memory device **400** including peripheral circuits **402**, according to some aspects of the present disclosure. NAND Flash memory device **400** may be one example of flash memory **302** in FIG. 3. NAND Flash memory device **400** can include a memory cell array **401** and peripheral circuits **402** operatively coupled to memory cell array **401**. Memory cells **406** in memory cell array **401** are provided in the form of an array of memory strings **408** each extending vertically above a substrate (not shown). In some implementations, each memory string **408** includes a plurality of memory cells **406** operatively coupled in series and stacked vertically. Each memory cell **406** can hold a continuous, analog value, such as an electrical voltage or charge, which depends on the number of electrons trapped within a region of memory cell **406**. Each memory cell **406** can be either a floating gate type

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of memory cell including a floating-gate transistor or a charge trap type of memory cell including a charge-trap transistor.

In some implementations, each memory cell **406** is a single-level cell (SLC) that has two possible levels (memory states) and thus, can store one bit of data. For example, the first state "0" can correspond to a first range of threshold voltages, and the second state "1" can correspond to a second range of threshold voltages. In some implementations, each memory cell **406** is an xLC that is capable of storing more than a single bit of data in more than four levels. For example, the xLC may store two bits per cell (a.k.a., multi-level cell (MLC)), three bits per cell (a.k.a., triple-level cell (TLC)), or four bits per cell (a.k.a., quad-level cell (QLC)). Each xLC can be programmed to assume a range of possible nominal storage values (i.e., corresponding to 2^N pieces of N-bits data). In some implementations, each memory cell **406** is set to one of 2^N levels corresponding to a piece of N-bits data, where N is an integer greater than 1. N may denote the total number of bits per cell. For example, N=2 for MLC, N=3 for TLC, or N=4 for QLC.

As shown in FIG. 4, each memory string **408** can also include a source select gate (SSG) transistor **410** at its source end and a drain select gate (DSG) transistor **412** at its drain end. SSG transistor **410** and DSG transistor **412** can be configured to activate select memory strings **408** (columns of the array) during read and program operations. In some implementations, the sources of memory strings **408** in the same block **404** are coupled through a same source line (SL) **414**, e.g., a common SL. In other words, all memory strings **408** in the same block **404** have an array common source (ACS), according to some implementations. The drain of each memory string **408** is coupled to a respective bit line **416** from which data can be read or written via an output bus (not shown), according to some implementations. In some implementations, each memory string **408** is configured to be selected or deselected by applying a select voltage or a deselect voltage to the gate of respective DSG transistor **412** through one or more DSG lines **413** and/or by applying a select voltage or a deselect voltage to the gate of respective SSG transistor **410** through one or more SSG lines **415**.

As shown in FIG. 4, memory strings **408** can be organized into multiple blocks **404**, each of which can have a common source line **414**, e.g., coupled to the ACS. In some implementations, each block **404** is the basic data unit for erase operations, i.e., all memory cells **406** on the same block **404** are erased at the same time. To erase memory cells **406** in a select block **404**, source lines **414** coupled to select block **404** as well as unselect blocks **404** in the same plane as select block **404** can be biased with an erase voltage (Vers), such as a high positive bias voltage (e.g., 20 V or more).

Memory cells **406** of adjacent memory strings **408** can be coupled through word lines **418** that select which row of memory cells **406** is affected by read and program operations. In some implementations, each word line **418** is coupled to a physical page **420** of memory cells **406**, which is the basic data unit for read and write (program) operations. The size of one physical page **420** in bits can relate to the number of memory strings **408** coupled by word line **418** in one block **404**. Each word line **418** can include a plurality of control gates (gate electrodes) at each memory cell **406** in respective physical page **420** and a gate line coupling the control gates.

Peripheral circuits **402** can be operatively coupled to memory cell array **401** through bit lines **416**, word lines **418**, source lines **414**, SSG lines **415**, and DSG lines **413**. Peripheral circuits **402** can include any suitable analog,

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digital, and mixed-signal circuits for facilitating the operations of memory cell array **401** by applying and sensing voltage signals and/or current signals to and from each select memory cell **406** through bit lines **416**, word lines **418**, source lines **414**, SSG lines **415**, and DSG lines **413**. Peripheral circuits **402** can include various types of peripheral circuits formed using complementary metal-oxide-semiconductor (CMOS) technologies.

FIG. 5 illustrates a schematic circuit diagram of a DRAM device **500** including peripheral circuits **502**, according to some aspects of the present disclosure. DRAM device **500** may be one example of DRAM **304** in FIG. 3. DRAM device **500** can include a memory cell array **501** and peripheral circuits **502** operatively coupled to memory cell array **501**. Memory cells **503** can be arranged in memory cell array **501** having rows and columns. DRAM device **500** requires periodic refreshing of memory cells **503**. In some implementations, each memory cell **503** includes a capacitor **507** for storing a bit of data as a positive or negative electrical charge as well as a transistor **505** that controls access to capacitor **507**. That is, each memory cell **503** shown in FIG. 5 is a one-transistor, one-capacitor (1T1C) cell, according to some implementations.

DRAM device **500** can include word lines **504** coupling peripheral circuits **502** and memory cell array **501** for controlling the switch of transistors **505** in memory cells **503** located in a row, as well as bit lines **506** coupling peripheral circuits **502** and memory cell array **501** for sending data to and/or receiving data from memory cells **503** located in a column. That is, each word line **504** is coupled to a respective row of memory cells **503**, and each bit line **506** is coupled to a respective column of memory cells **503**. The gate of transistor **505** can be coupled to word line **504**, one of the source and the drain of transistor **505** can be coupled to bit line **506**, the other one of the source and the drain of transistor **505** can be coupled to one electrode of capacitor **507**, and the other electrode of capacitor **507** can be coupled to the ground.

Peripheral circuits **502** can be coupled to memory cell array **501** through bit lines **506**, word lines **504**, and any other suitable metal wirings. Peripheral circuits **502** can include any suitable circuits for facilitating the operations of memory cell array **501** by applying and sensing voltage signals and/or current signals through word lines **504** and bit lines **506** to and from each memory cell **503**. Peripheral circuits **502** can include various types of peripheral circuits formed using CMOS technologies.

FIG. 6 illustrates a detailed schematic diagram of a memory system **600** managing an L2P mapping table and performing data search based on the L2P mapping table, according to some aspects of the present disclosure. Memory system **600** may be one example of memory system **102** in FIG. 1. As shown in FIG. 6, memory system **600** can include a memory controller **601**, a volatile memory device **602**, and a non-volatile memory device **604**. Memory controller **601** may be one example of memory controller **106** in FIG. 1. Volatile memory device **602** and non-volatile memory device **604** may be examples of memory devices **104** in FIG. 1. In some implementations, volatile memory device **602** includes DRAM (e.g., DRAM device **500** in FIG. 5), and non-volatile memory device **604** includes NAND Flash memory (e.g., NAND Flash memory device **400** in FIG. 4). In some implementations, memory controller **601** is further configured to cache a piece of host/user data in volatile memory device **602** or flush the piece of host/user data from volatile memory device **602** to non-volatile memory device **604**.

To enable data search and access, non-volatile memory device **604** can be divided into multiple memory regions **605** each has a unique physical address. In some implementations, each memory region **605** includes one or more logical pages, for example, a portion (e.g., $\frac{1}{2}$, $\frac{1}{4}$, or $\frac{1}{8}$) of one physical page **420** of NAND Flash memory device **400**. For example, the size of each memory region **605** may be 4,096 bytes. It is understood that memory region **605** may correspond to any suitable memory cell groups in non-volatile memory device **604** besides pages, such as portions of a page, blocks (e.g., blocks **404** of NAND Flash memory device **400**), etc. For example, the physical address of memory region **605** can be referred to as a physical allocation address (PAA), and a logical address corresponding to the PAA can be referred to as a logical allocation address (LAA). In another example, the physical address of memory region **605** can be a physical page address (PPA) when memory region **605** corresponds to a page of non-volatile memory device **604**, and a logical address corresponding to the PPA can be a logical block address (LBA).

Consistent with the scope of the present disclosure, to enable data search and access, cache **606** of volatile memory device **602** can be divided into multiple memory blocks **607** each having a unique identifier (ID, a.k.a., memory block ID). In some implementations, each memory block **607** includes one or more pages, for example, rows or columns of memory cells **503** of DRAM device **500**. In some implementations, to enable uniform data search between non-volatile memory device **604** and volatile memory device **602**, the size of each memory region **605** and the size of each memory block **607** may be the same. It is understood that in some examples, the size of each memory region **605** and the size of each memory block **607** may be different. For example, the size of each memory block **607** may be 4,096 bytes as well. It is understood that memory block **607** may correspond to any suitable memory cell groups in volatile memory device **602** besides pages, such as portions of a page, codewords, etc.

Cache **606** can be a portion of volatile memory device **602** that temporarily stores (caches) the frequently used and/or recently accessed data to speed up the read and write operations of non-volatile memory device **604**. Any suitable caching algorithms can be used to determine which data should be stored in cache **606** and when it should be replaced, including, for example, least recently used (LRU), most recently used (MRU), and first-in, first-out (FIFO). In some implementations, data from the host (host/user data) is first cached in cache **606** of volatile memory device **602**, and flushed to non-volatile memory device **604** under certain conditions based on the caching algorithm. For example, when the size of the data in cache **606** reaches a preset threshold (maximum caching size), data in cache **606** may be flushed to non-volatile memory device **604**. Cache **606** can be implemented by any suitable type of volatile memory device **602**, for example, DRAM **304** and/or cache **310** in FIG. 3.

Consistent with the scope of the present disclosure, to enable uniform search and access of the data, a uniform, expanded L2P mapping table **612** can be maintained and stored in volatile memory device **602** to map the logical addresses of data, not only to the physical addresses **616** (e.g., PPAs) of memory regions **605** in non-volatile memory device **604**, respectively, but also to the IDs **614** of memory blocks **607** in cache **606** of volatile memory device **602**, respectively. The logical addresses can identify the host/user data and be known to memory controller **601**. In some implementations, a logical address indicates the basic logi-

cal unit of data for each read or write operation, such as a logical block address (LBA). In some implementations, to enable uniform data search between non-volatile memory device **604** and volatile memory device **602**, the size of each memory region **605**, the size of each memory block **607**, and the size of the data corresponding to each logical address may be the same. For example, the size of the data corresponding to each logical address may be 4,096 bytes as well. Since memory controller **601** operates based on logical addresses, as opposed to physical addresses (e.g., physical addresses **616** or IDs **614**), L2P mapping table **612** can be used to enable the conversion between logical addresses and physical addresses across both non-volatile memory device **604** and volatile memory device **602** in a uniform manner, as described below in detail.

In some implementations, L2P mapping table **612** can be stored in non-volatile memory device **604**. In some other implementations, L2P mapping table **612** can be stored in any suitable type of volatile memory device **602**, such as DRAM **304** in FIG. 3. For example, the same volatile memory device **602**, such as DRAM **304** in FIG. 3, includes both cache **606** and L2P mapping table **612**. It is understood that in some examples, cache **606** and L2P mapping table **612** may be included in different volatile memory devices **602**. For example, cache **606** may be included in an SRAM, while DRAM **304** may include L2P mapping table **612**. Although L2P mapping table **612** is shown in FIG. 6 as being outside of cache **606**, it is understood that in some examples, L2P mapping table **612** may be stored in cache **606** as well.

In some implementations, L2P mapping table **612** can be stored in volatile memory device **602** with the addresses in volatile memory device **602**. For example, as shown in FIG. 7, L2P mapping table **612** may include addresses **704** in volatile memory device **602** (VM Add) each associated with a value **706**. Values **706** may include two types of information: IDs **614** of memory blocks **607** in cache **606** (e.g., ID1, ID2, ID3, etc.), and physical addresses **616** (PPAs) of memory regions **605** in non-volatile memory device **604** (e.g., PPA1, PPA2, PPA3, PPA4, etc.). As shown in FIG. 7, L2P mapping table **612** may map logical addresses **702** (LBAs) of host/user data to IDs **614** of memory blocks **607** in cache **606** and physical addresses **616** of memory regions **605** in non-volatile memory device **604** through addresses **704**. For example, for each piece of host/user data, a corresponding address **704** for an entry of L2P mapping table **612** in volatile memory device **602** may be determined based on the respective LBA **702** associated with the piece of host/user data. In one example as shown in FIG. 7, a respective LBA **702** (e.g., 0, 1, 2, 3, 4, 5, 6, etc.) associated with an entry can be multiplied by an entry size A and then added with an address offset (OFF) to form a corresponding address **704** of the entry, where the entry size A may represent a length of value **706** stored in the entry (e.g., $\Delta=4$ bytes). The address offset can be determined, for example, based on where L2P mapping table **612** is stored in volatile memory device **602**. The corresponding value **706** at the determined address **704** of L2P mapping table **612** may thus be determined, which indicates either an ID **614** of a memory block **607** in cache **606** or a physical address **616** of a memory region in non-volatile memory device **604**. As such, LBAs **702** of host/user data may be mapped to a plurality of physical addresses (e.g., IDs **614** of memory blocks **607** in cache **606** and physical addresses **616** of memory regions **605** in non-volatile memory device **604**), respectively, by L2P mapping table **612**.

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Referring back to FIG. 6, memory controller 601 can include multiple I/O interfaces, including a volatile memory interface 620 operatively coupled to volatile memory device 602, a non-volatile memory interface 622 operatively coupled to non-volatile memory device 604, and a host interface 618 operatively coupled to cache 606 of volatile memory device 602 and the host (not shown). Examples of those I/O interfaces may include DRAM interface 314, flash interface 312, and host interface 316 in FIG. 3, which may implement any suitable communication protocols facilitating data transfer, communication, and management, such as the NVMe protocol and PCI-E protocol, DDR protocol, to name a few.

Host interface 618 can be configured to receive write requests and read requests from the host. Each write request can be indicative of a piece of data associated with a logical address (e.g., LBA) to be written to memory system 600. Similarly, each read request can be indicative of a piece of data associated with a logical address (e.g., LBA) to be read from memory system 600. In some implementations, in response to receiving a write request or a read request, host interface 618 is also configured to fetch the piece of data from the host to temporarily store (cache) the piece of data in cache 606, or vice versa. For example, host interface 618 may include a direct memory access (DMA) unit that accesses data from and to cache 606.

Non-volatile memory interface 622 can be configured to enable memory controller 601 to access data stored in non-volatile memory device 604 based on the physical addresses (e.g., PPAs) of memory regions 605. Volatile memory interface 620 can be configured to enable memory controller 601 to access data stored in volatile memory device 602, such as to manage L2P mapping table 612 and to access data in cache 606. In some implementations, volatile memory interface 620 is configured to convert IDs 614 of memory blocks 607 in cache 606 to physical addresses of volatile memory device 602 that can be used directly by memory controller 601 for operating the memory cells of volatile memory device 602. In other words, while IDs 614 of memory blocks 607 in cache 606 can be used to facilitate the data search by L2P mapping table 612, memory controller 601 can still use the physical addresses of volatile memory device 602 to access data in volatile memory device 602. As a result, volatile memory device 602 does not need to be modified to accommodate the usage of IDs 614 of memory blocks 607 for data search, according to some implementations.

As shown in FIG. 6, memory controller 601 can further include a range division accelerator 608 and one or more L2P search engines 610 operatively coupled to volatile memory device 602, non-volatile memory interface 622, and volatile memory interface 620. In some implementations, range division accelerator 608 and L2P search engines 610 are firmware modules implemented by firmware codes/instructions stored in a memory (e.g., ROM 311 in FIG. 3 or flash memory 302 in FIG. 3) and executed by a processing unit (e.g., processing unit 308 in FIG. 3). In some implementations, range division accelerator 608 and L2P search engines 610 are hardware modules implemented by dedicated circuits, such as ASICs, for performing their dedicated functions described herein. The hardware implementation of range division accelerator 608 and L2P search engines 610 can improve search parallelism and reduce firmware overhead, thereby further improving data search performance.

Range division accelerator 608 can be configured to generate data search requests based on the read and write requests from the host via host interface 618, and assign the

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search requests to L2P search engines 610. That is, range division accelerator 608 can divide the read requests or write requests into search requests to be handled by multiple L2P search engines 610 in parallel, for example, based on the different logical addresses associated with the data of the read requests or write requests. For example, for each search request, range division accelerator 608 may identify an idle L2P search engine 610 to handle the search request. In some implementations, in response to receiving a write request indicative of a piece of data associated with a logical address (e.g., LBA), range division accelerator 608 is configured to assign the piece of the data to one of memory blocks 607 in cache 606 with a unique one of IDs 614, which triggers host interface 618 to fetch the corresponding piece of data from the host to the corresponding memory block 607 in cache 606.

L2P search engines 610 can be configured to handle the search requests and maintain L2P mapping table 612 stored in volatile memory device 602 through volatile memory interface 620 based on the handling of the search requests. In some implementations, a single L2P mapping table 612 is maintained for memory system 600, and multiple L2P search engines 610 are configured to maintain the same L2P mapping table 612 and use the same L2P mapping table 612 for data search. For example, multiple L2P search engines 610 may be configured to search multiple pieces of data, respectively, in parallel based on the same L2P mapping table 612. It is understood that in some examples, a single L2P search engine 610 may be used to handle the search requests. In some implementations, in response to host interface 618 fetching a piece of data from the host to the corresponding memory block 607 in cache 606 in response to the write request, L2P search engine 610 may be configured to update L2P mapping table 612 to map the logical address (e.g., LBA) associated with the piece of data to the unique ID 614 of the corresponding memory block 607. For example, as shown in FIG. 7, assuming LBA 702 of the piece of data is "3," value 706 may be updated by search engine 610 to become "ID3," which is the unique ID 614 of the corresponding memory block 607. Value 706 may be stored at address 704 "OFF+3Δ" of L2P mapping table 612 in volatile memory device 602. In some implementations, in response to the piece of cached data in cache 606 being flushed to one of memory regions 605 in non-volatile memory device 604 with a unique one of physical addresses 616 (e.g., PPA), L2P search engine 610 is further configured to update L2P mapping table 612 to map the logical address (e.g., LBA) associated with the piece of data to the unique physical address 616 of the corresponding memory region 605. For example, as shown in FIG. 7, assuming LBA 702 of the piece of data is "3," value 706 may be updated by L2P search engine 610 to become "PPA5," which is the unique physical address 616 of the corresponding memory region 605, since the piece of data associated with LBA 3 has been moved from ID3 in cache 606 to PPA5 in non-volatile memory device 604.

In some implementations, in response to receiving a search request for a read request indicative of a piece of data with a logical address (e.g., LBA), L2P search engine 610 is configured to search the piece of data based on the logical address and L2P mapping table 612. L2P search engine 610 can be configured to determine an address of L2P mapping table 612 in volatile memory device 602 based on the logical address, and then determine the value at the address of L2P mapping table 612. L2P search engine 610 may identify whether the value is an ID 614 of memory block 607 in cache 606 or a physical address 616 of memory region 605

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in non-volatile memory device **604** by performing operations like those described below with reference to FIGS. **11A** and **11B**.

In one example, as shown in FIG. **7**, assuming LBA **702** of the piece of data is "0," L2P search engine **610** may first add LBA **702** "0" multiplied by the entry size A to the address offset "OFF" to obtain address **704** "OFF," and then identify value **706** at address **704** "OFF" to be "ID1," meaning that the piece of data to be read is data currently cached in memory block ID1 in cache **606**. In another example, as shown in FIG. **7**, assuming LBA **702** of the piece of data is "2," L2P search engine **610** may first add LBA **702** "2" multiplied by the entry size A to the address offset "OFF" to obtain address **704** "OFF+2A," and then identify value **706** at address **704** "OFF+2A" to be "PPA1," meaning that the piece of data to be read is data currently stored in memory region PPA1 in non-volatile memory device **604**.

In some implementations, in response to identifying the ID **614** of memory block **607** in cache **606**, L2P search engine **610** provides the identified ID **614** to volatile memory interface **620**, and volatile memory interface **620** converts the ID **614** to a corresponding physical address in volatile memory device **602**, such that host interface **618** can fetch the piece of data from the corresponding physical address in volatile memory device **602**, for example, using DMA. In some implementations, in response to identifying the physical address **616** of memory region **605** in non-volatile memory device **604**, L2P search engine **610** provides the identified physical address **616** (e.g., PPA) to non-volatile memory interface **622**, such that non-volatile memory interface **622** can fetch the piece of data from the corresponding physical address in non-volatile memory device **604**.

FIG. **8** illustrates a flowchart of a method **800** for operating a memory controller, according to some aspects of the present disclosure. The memory controller may be any suitable memory controller disclosed herein, such as memory controller **601**. It is understood that the operations shown in method **800** may not be exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. **8**.

The memory controller is operatively coupled to a volatile memory device and a non-volatile memory device. The volatile memory device can include a cache. The cache is divided into memory blocks each has a respective unique one of IDs. The non-volatile memory device is divided into memory regions each having a respective unique one of physical addresses. For example, as shown in FIG. **6**, volatile memory device **602** may include cache **606**, which may be divided into memory blocks **607** each having a unique ID **614**, and non-volatile memory device **604** may be divided into memory regions **605** each having a unique physical address **616**.

Referring to FIG. **8**, method **800** starts at operation **802**, in which an L2P mapping table is generated and stored in the volatile memory device. For example, as shown in FIG. **6**, memory controller **601** may generate L2P mapping table **612** during the initiation of memory system **600** and store L2P mapping table **612** in volatile memory device **602** during the operation of memory system **600**. For example, as shown in FIG. **7**, L2P mapping table **612** may include values **706** at each address **704** thereof, which may be mapped to LBAs **702**, respectively.

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Method **800** proceeds to operation **804**, as illustrated in FIG. **8**, in which a first set of data is cached in the memory blocks of the cache. The first set of data is associated with a first set of logical addresses, respectively. For example, as shown in FIG. **6**, memory controller **601** may cache pieces of host/user data in memory blocks **607**, respectively, in cache **606**. Each piece of host/user data may be associated with a respective one of logical addresses (e.g., LBAs). In one example, host interface **618** of memory controller **601** may fetch host/user data from the host to cache the host/user data in memory blocks **607** using DMA.

Method **800** proceeds to operation **806**, as illustrated in FIG. **8**, in which the L2P mapping table is maintained, such that the L2P mapping table maps the first set of logical addresses to the IDs of the memory blocks of the cache, respectively. For example, as shown in FIGS. **6** and **7**, memory controller **601** may maintain L2P mapping table **612**, such that a set of LBAs **702** may be mapped to IDs **614** of memory blocks **607** via addresses **704**, respectively.

Method **800** proceeds to operation **808**, as illustrated in FIG. **8**, in which a second set of data is stored in the memory regions of the non-volatile memory device. The second set of data is associated with a second set of logical addresses, respectively. For example, as shown in FIG. **6**, memory controller **601** may store pieces of host/user data in memory regions **605**, respectively, in non-volatile memory device **604**. Each piece of host/user data may be associated with a respective one of logical addresses (e.g., LBAs). In one example, the pieces of host/user data may be flushed from cache **606** to non-volatile memory device **604**.

Method **800** proceeds to operation **810**, as illustrated in FIG. **8**, in which the L2P mapping table is maintained, such that the L2P mapping table maps the second set of logical addresses to the physical addresses of the memory regions of the non-volatile memory device, respectively. For example, as shown in FIGS. **6** and **7**, memory controller **601** may maintain L2P mapping table **612**, such that another set of LBAs **702** may be mapped to physical addresses **616** (e.g., PPAs) of memory regions **605** via addresses **704**, respectively.

Method **800** proceeds to operation **812**, as illustrated in FIG. **8**, in which a piece of data is searched based on the L2P mapping table. As shown in FIG. **6**, multiple L2P search engines **610** of memory controller **601** may perform data search of multiple pieces of data in parallel based on the logical addresses of the multiple pieces of data and L2P mapping table **612**.

In some implementations, to search the piece of data, an address of the L2P mapping table in the volatile memory device is determined based on a logical address associated with the piece of the data, and a value at the address of the L2P mapping table is determined. For example, as shown in FIGS. **6** and **7**, each L2P search engine **610** may calculate address **704** of L2P mapping table **612** based on LBA **702** and the address offset, and identify value **706** at address **704** to be either an ID of a memory block of the volatile memory device or a physical address (e.g., PPA) of a memory region of the non-volatile memory device. In some implementations, to search the piece of data, in response to the value being one of the IDs of the memory blocks in the volatile memory device, the piece of the data is fetched from the memory block in the volatile memory device having the ID. In some implementations, to search the piece of data, in response to the value being one of the physical addresses of the memory regions in the non-volatile memory device, the piece of the data is fetched from the memory region in the non-volatile memory device having the physical address.

For example, as shown in FIGS. 6 and 7, if L2P search engine 610 identifies value 706 at address 704 to be an ID 614, host interface 618 may fetch the piece of data from memory block 607 in cache 606 that has the identified ID 614, for example, using DMA. In contrast, if L2P search engine 610 identifies value 706 at address 704 to a physical address 616 (e.g., PPA), non-volatile memory interface 622 may fetch the piece of data from memory region 605 in non-volatile memory device 604 that has the identified physical address 616.

In some implementations, a write request may be indicative of a first piece of data from the first set of data associated with the first set of logical addresses, where the first piece of data is associated with a first logical address from the first set of logical addresses. In response to receiving the write request, the first piece of data is assigned to a first memory block of the memory blocks having a first ID of the IDs (e.g., the first piece of data is fetched and cached into the first memory block). For example, as shown in FIG. 9, at 902, write requests may be divided by range division accelerator 608 into multiple search requests each associated with a piece of host/user data and a logical address thereof. At 904, memory block IDs may be assigned by range division accelerator 608 to the pieces of host/user data, respectively, such that each piece of host/user data may be associated with a respective memory block ID. At 906, range division accelerator 608 may trigger host interface 618 to fetch each piece of host/user data to the respective memory block using DMA.

In some implementations, in response to fetching the first piece of data to the first memory block, the L2P mapping table is updated to map the first logical address to the first ID. In response to the first piece of data is flushed from the first memory block of the cache to a memory region of the non-volatile memory device having a physical address, the L2P mapping table is updated to map the first logical address to the physical address. For example, at 908 of FIG. 9, the memory block ID where the piece of host/user data is cached may be updated by L2P search engine 610 in L2P mapping table 612 to be mapped to the logical address of the piece of host/user data. At 910, whether the non-volatile memory device programming is done may be checked. The non-volatile memory device programming may be performed by flushing the cached host/user data from the cache to the non-volatile memory device. Once the non-volatile memory device programming is done, each piece of host/user data may be stored in a respective one of memory regions each associated with a PPA. If the answer to 910 is yes, at 912, the PPA where the piece of host/user data is stored in the non-volatile memory device may be updated by L2P search engine 610 in L2P mapping table 612 to be mapped to the logical address of the piece of host/user data, replacing the memory block ID. In some implementations, the update of the L2P mapping table with the PPA may be performed before the non-volatile memory device programming is done. For example, the update of the L2P mapping table with the PPA can be performed when a command to program the piece of host/user data into the non-volatile memory device is generated, e.g., before the programming is actually done. At 914, whether new incoming write requests are received by memory controller (e.g., memory controller 300 in FIG. 3) from host (e.g., host 306 in FIG. 3) may be checked to determine whether the process may continue from 902 again for the new incoming write requests. If the answer to 910 is no, the process may proceed to 914 directly, bypassing 912 without updating L2P mapping table 612.

In some implementations, a read request may be indicative of a second piece of data from the first set of data associated with the first set of logical addresses, where the second piece of data is associated with a second logical address from the first set of logical addresses. In response to receiving the read request, an address of the L2P mapping table in the volatile memory device can be determined based on the second logical address, and a second ID of the second memory block is identified at the address of the L2P mapping table in the volatile memory device. For example, as shown in FIG. 10, at 1002, read requests may be divided by range division accelerator 608 into multiple search requests each associated with a logical address. At 1004, idle L2P search engines 610 may be identified by range division accelerator 608 to perform the search requests. At 1006, the DRAM address may be located by L2P search engine 610 based on the logical address, e.g., by multiplying the logical address with an entry size A to form a product, and then adding the product to the address offset to obtain the DRAM address. At 1008, the value at the DRAM address may be fetched by L2P search engine 610.

In some implementations, in response to the value being one of the IDs of the memory blocks in the volatile memory device, the second piece of data is fetched from the second memory block of the memory blocks in the cache based on the L2P mapping table. For example, as shown in FIG. 10, at 1010, whether the piece of host/user data is in the non-volatile memory device or not may be determined based on the fetched value (either a memory block ID or a PPA, fetched at 1008). If the answer to 1010 is no, meaning that the piece of host/user data is still in the cache, at 1014, the piece of host/user data may be fetched by volatile memory interface 620 from the cache based on the fetched memory block ID at the DRAM address. If the answer to 1010 is yes, meaning that the piece of host/user data is in the non-volatile memory device, at 1012, the piece of host/user data may be read by non-volatile memory interface 622 from the non-volatile memory device based on the fetched PPA at the DRAM address. In either case, at 1016, the piece of host/user data may be transmitted to the host by host interface 618.

FIG. 11A illustrates an example approach to organize physical addresses mapped by an L2P mapping table, according to some aspects of the present disclosure. As described above, the L2P mapping table may be used to map a plurality of logical addresses to a plurality of physical addresses, respectively. In the example approach of FIG. 11A, the plurality of physical addresses may be classified into either a first type of volatile memory addresses (e.g., IDs of memory blocks of a volatile memory device such as a DRAM) or a second type of non-volatile memory addresses (e.g., physical addresses of memory regions of a non-volatile memory device such as PPAs of a NAND memory device). In order to distinguish between the first type of volatile memory addresses and the second type of non-volatile memory addresses, at least one bit of an address data width can be reserved for marking the types of the physical addresses.

For example, as shown in FIG. 11A, each physical address may have an address data width of 32 bits. The most significant bit of the 32 bits can be reserved and used to mark a type of the physical address, and the remaining 31 bits can be used to express a specific value of the physical address. For example, the first type of volatile memory addresses can be identified by marking the most significant bit of each address to be "0," and the second type of non-volatile

memory addresses can be identified by marking the most significant bit of each address to be “1.”

By reserving the most significant bit in each address for the marking purpose, the possible size of the physical space that the address data width can express is reduced from a range of $(0, 2^{32}-1)$ to a range of $(0, 2^{31}-1)$. For an SSD (e.g., enterprise SSD) with a large storage capacity, the remaining bits of the address data width may not be sufficient to express all the physical addresses of the SSD. To improve the usage efficiency of the address data width, another example approach to organize the plurality of physical addresses mapped by the L2P mapping table is disclosed herein with combined reference to FIGS. 3, 6, and 11B.

In the example approach of FIG. 11B, the plurality of physical addresses mapped by the L2P mapping table are organized in a way different from that in the example approach of FIG. 11A. Specifically, L2P search engine 610 shown in FIG. 6 may be configured to organize the plurality of physical addresses into a plurality of address categories based on at least one address boundary. Referring to FIG. 11A, the plurality of address categories may include at least one of: (1) a first category of user data addresses 1164 mapping to memory regions of a user area 1172 of a non-volatile memory device; (2) a second category of volatile memory addresses 1162 mapping to memory blocks of a volatile memory device; or (3) a third category of specialized memory addresses 1160 mapping to memory regions of a system area 1170 of the non-volatile memory device.

Each specialized memory address 1160 in the third category may be dedicated to a particular purpose and can be mapped to a memory region in system area 1170 of the non-volatile memory device. In some implementations, a specialized memory address 1160 may be used to indicate relevant information (e.g., validity) related to data stored in physical addresses. For example, assuming that a piece of user data is already stored in a particular physical address of user area 1172 such as PPA2 (e.g., the L2P mapping table is already updated to map a first logical address to PPA2). Then, upon receiving an instruction from a host to delete the piece of user data, the memory controller may modify the L2P mapping table to map the first logical address to a specialized memory address 1160 which is set to be a particular physical address of a memory region in system area 1170 such as PPA1. In this case, specialized memory address 1160 being set to PPA1 can be used to mark the invalidity of the piece of user data associated with the first logical address. The content or the value of an entry mapping to the first logical address in the L2P mapping table is set to be the particular physical address of the memory region in system area 1170 (e.g., PPA1), and the piece of user data associated with the first logical address is identified to be invalid.

In some implementations, the at least one address boundary may include a first address boundary 1168 and a second address boundary 1166 stored in registers 330 and 332, respectively. Second address boundary 1166 and/or first address boundary 1168 may be used to determine an address category of each physical address, and can be predetermined or configured by the memory controller. Each specialized memory address 1160 in the third category can be smaller than second address boundary 1166 (e.g., specialized memory address $1160 < \text{second address boundary } 1166$). Each user data address 1164 in the first category can be greater than first address boundary 1168 (e.g., user data address $1164 \geq \text{first address boundary } 1168$). Each volatile memory address 1162 in the second category can be greater than or equal to second address boundary 1166 and smaller

than or equal to first address boundary 1168 (e.g., second address boundary $1166 \leq \text{volatile memory address } 1162 \leq \text{first address boundary } 1168$).

In some implementations, the L2P mapping table may map a first set of logical addresses in the plurality of logical addresses to a first set of physical addresses associated with the memory regions of user area 1172 of the non-volatile memory device, respectively. Then, the first category of user data addresses 1164 may include the first set of physical addresses associated with the memory regions of user area 1172 of the non-volatile memory device.

The L2P mapping table may also map a second set of logical addresses in the plurality of logical addresses to IDs of the memory blocks of the volatile memory device, respectively. Then, the second category of volatile memory addresses 1162 may include the IDs of the memory blocks of the volatile memory device.

The L2P mapping table may map a third set of logical addresses in the plurality of logical addresses to a third set of physical addresses associated with the memory regions of system area 1170 of the non-volatile memory device, respectively. Then, the third category of specialized memory addresses 1160 may include the third set of physical addresses associated with the memory regions of system area 1170 of the non-volatile memory device.

In some implementations, range division accelerator 608 of FIG. 6 may generate a search request based on a read request from a host via host interface 618, and assign the search request to L2P search engine 610. The read request may be indicative of retrieving a piece of data associated with a logical address. In response to receiving the search request for the read request, L2P search engine 610 may be configured to determine an address of an entry in the L2P mapping table based on the logical address (e.g., the address of the entry = the logical address $\times \Delta + \text{OFF}$, as shown in FIG. 7). L2P search engine 610 may identify a physical address stored in the entry of the L2P mapping table based on the address of the entry. For example, the identified physical address can be the content or the value stored in the entry of the L2P mapping table.

Next, L2P search engine 610 may determine an address category which the physical address is classified into based on at least one of first address boundary 1168 or second address boundary 1166. For example, responsive to the physical address being smaller than second address boundary 1166, L2P search engine 610 may determine that the physical address is classified into the third category of specialized memory addresses 1160. Or, responsive to the physical address being greater than first address boundary 1168, L2P search engine 610 may determine that the physical address is classified into the first category of user data addresses 1164. Or, responsive to the physical address being equal to or greater than second address boundary 1166 and being equal to or smaller than first address boundary 1168, L2P search engine 610 may determine that the physical address is classified into the second category of volatile memory addresses 1162.

Then, L2P search engine 610 may instruct to fetch the piece of data from one of the volatile memory device and the non-volatile memory device based on the determined address category. Specifically, if the physical address is classified into the third category of specialized memory addresses 1160 or the first category of user data addresses 1164, L2P search engine 610 may instruct to fetch the piece of data from the non-volatile memory device using the physical address. For example, L2P search engine 610 may provide the physical address (e.g., PPA2) to non-volatile

memory interface **622**, such that non-volatile memory interface **622** can fetch the piece of data from the corresponding physical address in the non-volatile memory device. Or, if the physical address is classified into the second category of volatile memory addresses **1162**, L2P search engine **610** may instruct to fetch the piece of data from the volatile memory device using the physical address. For example, L2P search engine **610** can provide the physical address (e.g., an identified ID) to volatile memory interface **620**, such that host interface **618** can fetch the piece of data from the corresponding physical address in volatile memory device **602**, for example, using DMA.

The example approach shown in FIG. **11B** does not reserve any bit in the address data width for marking the different types or categories of the addresses or other purpose for the data, such that all bits in the address data width (e.g., all 32 bits) can be used to express the physical addresses of the physical space corresponding to the non-volatile memory device or the volatile memory device. Alternatively, since there is no reserving bit in the example approach of FIG. **11B**, a reduced address data width (e.g., 31 bits rather than 32 bits) can be used to express the physical addresses of the physical space corresponding to the non-volatile memory device. In this case, the size of the L2P mapping table can be reduced since the physical addresses stored in the L2P mapping table are reduced to 31 bits. Therefore, the cost of the DRAM where the L2P mapping table is stored can be reduced.

FIG. **12** illustrates a flowchart of another method **1200** for operating a memory controller, according to some aspects of the present disclosure. The memory controller may be any suitable memory controller disclosed herein, such as memory controller **601**. It is understood that the operations shown in method **1200** may not be exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. **12**.

Referring to FIG. **12**, method **1200** starts at operation **1202**, in which an L2P mapping table is maintained, which maps logical addresses to physical addresses, respectively.

Method **1200** proceeds to operation **1204**, as illustrated in FIG. **12**, in which the physical addresses mapped by the L2P mapping table are organized into address categories based on at least one address boundary. For example, the at least one address boundary may include a first address boundary and a second address boundary. The plurality of physical addresses are organized into the following address categories based on the first address boundary and the second address boundary: (1) a first category of user data addresses mapping to memory regions of a user area of a non-volatile memory device; (2) a second category of volatile memory addresses mapping to memory blocks of a volatile memory device; or (3) a third category of specialized memory addresses mapping to memory regions of a system area of the non-volatile memory device.

FIG. **13A** illustrates a flowchart of an implementation of a method **1300** for handling a read request when physical addresses mapped by an L2P mapping table are organized using the example approach of FIG. **11A**, according to some aspects of the present disclosure. Method **1300** may be performed by a memory controller, which may be any suitable memory controller disclosed herein, such as memory controller **601**. It is understood that the operations shown in method **1300** may not be exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of

the operations may be performed simultaneously, or in a different order than shown in FIG. **13A**.

The memory controller is operatively coupled to a volatile memory device and a non-volatile memory device. The volatile memory device can include a cache. The cache is divided into memory blocks each has a respective unique one of IDs. The non-volatile memory device is divided into memory regions each having a respective unique one of physical addresses. For example, as shown in FIG. **6**, volatile memory device **602** may include cache **606**, which may be divided into memory blocks **607** each having a unique ID **614**, and non-volatile memory device **604** may be divided into memory regions **605** each having a unique physical address **616**.

Referring to FIG. **13A**, method **1300** starts at operation **1302**, in which read requests may be divided by range division accelerator **608** into multiple search requests each associated with a logical address.

Method **1300** proceeds to operation **1304**, as illustrated in FIG. **13A**, in which idle L2P search engines **610** may be identified by range division accelerator **608** to perform the search requests. Then, the following operations **1306-1316** may be performed for each search request.

Specifically, for a particular search request associated with a logical address, method **1300** proceeds to operation **1306**, as illustrated in FIG. **13A**, in which an L2P mapping table is searched to fetch a value (e.g., a physical address) stored in the L2P mapping table based on the logical address. For example, a DRAM address may be located by L2P search engine **610** based on the logical address, e.g., by multiplying the logical address with an entry size **A** to form a product, and then adding the product to an address offset of the L2P mapping table to obtain the DRAM address. Then, the value at the DRAM address may be fetched by L2P search engine **610**.

Method **1300** proceeds to operation **1308**, as illustrated in FIG. **13A**, in which the value (e.g., the physical address) is parsed by L2P search engine **610**. For example, L2P search engine **610** extracts the most significant bit of the physical address from the parsed value.

Method **1300** proceeds to operation **1310**, as illustrated in FIG. **13A**, in which it is determined whether the most significant bit of the physical address is 0. If the most significant bit of the physical address is 0, then method **1300** may proceed to operation **1312**. Otherwise (e.g., the most significant bit is 1), method **1300** may proceed to operation **1314**.

At operation **1312**, it is determined that the physical address represents a memory block ID of the volatile memory device since the most significant bit is 0. At operation **1313**, L2P search engine **610** may instruct to fetch a piece of data associated with the memory block ID from the volatile memory device. For example, the piece of data may be fetched by volatile memory interface **620** from the cache based on the memory block ID. At operation **1316**, the piece of data can be transmitted to a host by host interface **618**.

On the other hand, at operation **1314**, it is determined that the physical address represents a PPA in the non-volatile memory device since the most significant bit is 1. At operation **1315**, L2P search engine **610** may instruct to read the piece of data from the non-volatile memory device. For example, the piece of data may be read by non-volatile memory interface **622** from a NAND Flash memory device based on the PPA. The piece of data can also be transmitted to the host by host interface **618** at operation **1316**.

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FIG. 13B illustrates a flowchart of an implementation of another method 1350 for handling a read request when physical addresses mapped by an L2P mapping table are organized using the example approach of FIG. 11B, according to some aspects of the present disclosure. Method 1350 may be performed by a memory controller, which may be any suitable memory controller disclosed herein, such as memory controller 601. It is understood that the operations shown in method 1350 may not be exhaustive and that other operations can be performed as well before, after, or between any of the illustrated operations. Further, some of the operations may be performed simultaneously, or in a different order than shown in FIG. 13B.

Method 1350 may include operations like those of method 1300 of FIG. 13A, and the description of the similar operations will not be repeated herein.

At operation 1308 of method 1350, the value (e.g., the physical address) is parsed by L2P search engine 610. For example, L2P search engine 610 may compare the value (e.g., the physical address) with at least one of a first address boundary or a second address boundary.

At operation 1311 of method 1350, L2P search engine 610 may determine whether the value (e.g., the physical address) is equal to or greater than the second address boundary and equal to or smaller than the first address boundary. If the value (e.g., the physical address) is equal to or greater than the second address boundary and equal to or smaller than the first address boundary, method 1350 may proceed to operation 1312 to determine that the value (e.g., the physical address) represents a memory block ID of the volatile memory device. Otherwise (e.g., the value is smaller than the second address boundary or greater than the first address boundary), method 1350 may proceed to operation 1314 to determine that the value (e.g., the physical address) represents a PPA in the non-volatile memory device.

In various aspects of the present disclosure, the functions described herein may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored as instructions on a non-transitory computer-readable medium. Computer-readable media includes computer storage media. Storage media may be any available media that can be accessed by a memory controller, such as memory controller 601 in FIG. 6. By way of example, and not limitation, such computer-readable media can include RAM, ROM, electrically erasable programmable ROM (EEPROM), compact disc read-only memory (CD-ROM) or other optical disk storage, hard disk drive (HDD), such as magnetic disk storage or other magnetic storage devices, Flash drive, SSD, or any other medium that can be used to carry or store desired program code in the form of instructions or data structures and that can be accessed by a processing system, such as a mobile device or a computer. Disk and disc, as used herein, includes CD, laser disc, optical disc, digital video disc (DVD), and floppy disk where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above should also be included within the scope of computer-readable media.

The foregoing description of the specific implementations can be readily modified and/or adapted for various applications. Therefore, such adaptations and modifications are intended to be within the meaning and range of equivalents of the disclosed implementations, based on the teaching and guidance presented herein.

The breadth and scope of the present disclosure should not be limited by any of the above-described example

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implementations, but should be defined only in accordance with the following claims and their equivalents.

Although specific configurations and arrangements are discussed, it should be understood that this is done for illustrative purposes only. As such, other configurations and arrangements can be used without departing from the scope of the present disclosure. Also, the subject matter as described in the present disclosure can also be used in a variety of other applications. Functional and structural features as described in the present disclosures can be combined, adjusted, modified, and rearranged with one another and in ways that are consistent with the scope of the present disclosure.

What is claimed is:

1. A memory controller, comprising:

a logical-to-physical (L2P) search engine configured to: maintain an L2P mapping table that maps logical addresses to physical addresses, respectively; and compare the physical addresses mapped by the L2P mapping table with at least one address boundary to organize the physical addresses into address categories comprising a first category of user data addresses associated with a user area of a non-volatile memory device, a second category of volatile memory addresses associated with a volatile memory device, and a third category of specialized memory addresses associated with a system area of the non-volatile memory device,

wherein when one or more first logical addresses of the logical addresses are mapped to a specialized address in the third category of specialized addresses, data associated with the one or more first logical addresses is marked as invalid.

2. The memory controller of claim 1, wherein:

the at least one address boundary comprises a first address boundary; and

the memory controller further comprises a first register configured to store the first address boundary.

3. The memory controller of claim 2, wherein the first category of user data addresses maps to memory regions of the user area of the non-volatile memory device, and the second category of volatile memory addresses maps to memory blocks of the volatile memory device.

4. The memory controller of claim 3, wherein:

the L2P mapping table maps a first set of logical addresses in the logical addresses to a first set of physical addresses associated with the memory regions of the user area of the non-volatile memory device, respectively, and the first category of user data addresses comprises the first set of physical addresses associated with the memory regions of the user area of the non-volatile memory device; and

the L2P mapping table also maps a second set of logical addresses in the logical addresses to identifiers (IDs) of the memory blocks of the volatile memory device, respectively, and the second category of volatile memory addresses comprises the IDs of the memory blocks of the volatile memory device.

5. The memory controller of claim 3, wherein:

each user data address in the first category is greater than the first address boundary; and

each volatile memory address in the second category is smaller than the first address boundary.

6. The memory controller of claim 3, wherein:

the at least one address boundary further comprises a second address boundary lower than the first address boundary; and

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the memory controller further comprises a second register configured to store the second address boundary.

7. The memory controller of claim 6, wherein:

the third category of specialized memory addresses maps to memory regions of the system area of the non-volatile memory device; and

the L2P mapping table also maps a third set of logical addresses in the logical addresses to a third set of physical addresses associated with the memory regions of the system area of the non-volatile memory device, respectively, and the third category of specialized memory addresses comprises the third set of physical addresses associated with the memory regions of the system area of the non-volatile memory device.

8. The memory controller of claim 7, wherein each volatile memory address in the second category is greater than or equal to the second address boundary and smaller than or equal to the first address boundary; and

each specialized memory address in the third category is smaller than the second address boundary.

9. The memory controller of claim 3, further comprising a volatile memory device interface operatively coupled to the volatile memory device and a non-volatile memory device interface operatively coupled to the non-volatile memory device.

10. The memory controller of claim 3, wherein the volatile memory device comprises dynamic random-access memory (DRAM), and the non-volatile memory device comprises NAND Flash memory.

11. The memory controller of claim 8, wherein responsive to a read request indicative of retrieving a piece of data associated with a second logical address, the L2P search engine is further configured to:

determine an address of an entry in the L2P mapping table based on the second logical address;

identify a physical address stored in the entry of the L2P mapping table based on the address of the entry;

determine an address category which the physical address is classified into based on the first address boundary and the second address boundary; and

instruct to fetch the piece of data from one of the volatile memory device and the non-volatile memory device based on the address category.

12. The memory controller of claim 11, wherein to determine the address category, the L2P search engine is further configured to:

responsive to the physical address being greater than the first address boundary, determine that the physical address is classified into the first category of user data addresses mapping to the memory regions of the user area of the non-volatile memory device;

responsive to the physical address being lower than the second address boundary, determine that the physical address is classified into the third category of specialized memory addresses mapping to the memory regions of the system area of the non-volatile memory device; or

responsive to the physical address being equal to or greater than the second address boundary and being equal to or smaller than the first address boundary, determine that the physical address is classified into the second category of volatile memory addresses mapping to the memory blocks of the volatile memory device.

13. The memory controller of claim 12, wherein to instruct to fetch the piece of data, the L2P search engine is further configured to:

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responsive to the physical address being classified into the third category of specialized memory addresses or the first category of user data addresses, instruct to read the piece of data from the non-volatile memory device using the physical address; or

responsive to the physical address being classified into the second category of volatile memory addresses, instruct to fetch the piece of data from the volatile memory device using the physical address.

14. A memory system, comprising:

a non-volatile memory device comprising memory regions each associated with a physical address;

a memory controller operatively coupled to the non-volatile memory device and configured to control the non-volatile memory device, the memory controller comprising a logical-to-physical (L2P) search engine configured to:

maintain an L2P mapping table that maps logical addresses to physical addresses, respectively; and

compare the physical addresses mapped by the L2P mapping table with at least one address boundary to organize the physical addresses into address categories comprising a first category of user data addresses associated with a user area of a non-volatile memory device, a second category of volatile memory addresses associated with a volatile memory device, and a third category of specialized memory addresses associated with a system area of the non-volatile memory device,

wherein when one or more logical addresses of the logical addresses are mapped to a specialized address in the third category of specialized addresses, data associated with the one or more logical addresses is marked as invalid.

15. The memory system of claim 14, wherein:

the at least one address boundary comprises a first address boundary; and

the memory controller further comprises a first register configured to store the first address boundary.

16. The memory system of claim 15, wherein the first category of user data addresses maps to memory regions of the user area of the non-volatile memory device, and the second category of volatile memory addresses maps to memory blocks of the volatile memory device.

17. A method for operating a memory controller, comprising:

maintaining a logical-to-physical (L2P) mapping table that maps logical addresses to physical addresses, respectively; and

comparing the physical addresses mapped by the L2P mapping table with at least one address boundary to organize the physical addresses into address categories comprising a first category of user data addresses associated with a user area of a non-volatile memory device, a second category of volatile memory addresses associated with a volatile memory device, and a third category of specialized memory addresses associated with a system area of the non-volatile memory device, wherein when one or more logical addresses of the logical addresses are mapped to a specialized address in the third category of specialized addresses, data associated with the one or more logical addresses is marked as invalid.

18. The method of claim 17, wherein the at least one address boundary comprise a first address boundary stored in a first register.

19. The method of claim 18, wherein the first category of user data addresses maps to memory regions of the user area of the non-volatile memory device, and the second category of volatile memory addresses maps to memory blocks of the volatile memory device. 5

20. The method of claim 19, wherein:

the L2P mapping table maps a first set of logical addresses in the logical addresses to a first set of physical addresses associated with the memory regions of the user area of the non-volatile memory device, respectively, and the first category of user data addresses comprises the first set of physical addresses associated with the memory regions of the user area of the non-volatile memory device; and 10

the L2P mapping table also maps a second set of logical addresses in the logical addresses to identifiers (IDs) of the memory blocks of the volatile memory device, respectively, and the second category of volatile memory addresses comprises the IDs of the memory blocks of the volatile memory device. 15 20

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